

# Renovascular Hypertension and Ischemic Nephropathy

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## CHAPTER OUTLINE

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 NEPHROPATHY, 1366  
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## KEY POINTS

- Although poststenotic kidneys may lose glomerular filtration rate and tissue volume, the contralateral kidney without stenosis undergoes compensatory hypertrophy. These changes mask the damage to the affected kidney, making overall glomerular filtration rate an unreliable marker of the severity and progression of atherosclerotic renal artery stenosis.
- Most human renovascular hypertension is asymmetric. The contralateral, nonstenotic kidney is subjected to elevated systemic perfusion pressures. Rising perfusion pressure forces natriuresis from the nonstenotic kidney and suppresses renin release. Hence the nonstenotic kidney tends to prevent the rise in systemic pressures, thereby perpetuating reduced perfusion to the stenotic side and promoting continued renin release from the stenotic kidney. Hypertension is angiotensin dependent and is associated with elevated circulating levels of plasma renin activity.
- When no contralateral kidney is present or able to respond to pressure natriuresis, the mechanisms sustaining hypertension differ. Although renin release occurs initially, elevated systemic pressures develop as a result of sodium and volume retention due to ineffective natriuresis. Rising pressures eventually restore renin levels to normal.
- Management of cardiovascular risk and hypertension are the primary objectives of medical therapy. For most patients, the realistic goals of renal revascularization are to reduce blood pressure medication requirements and to stabilize renal function over time.
- The clinician must evaluate both the impact of the renal artery disease in the individual patient and the potential risk/benefit ratio for renal revascularization. In many cases intervention may not yield the desired outcome.

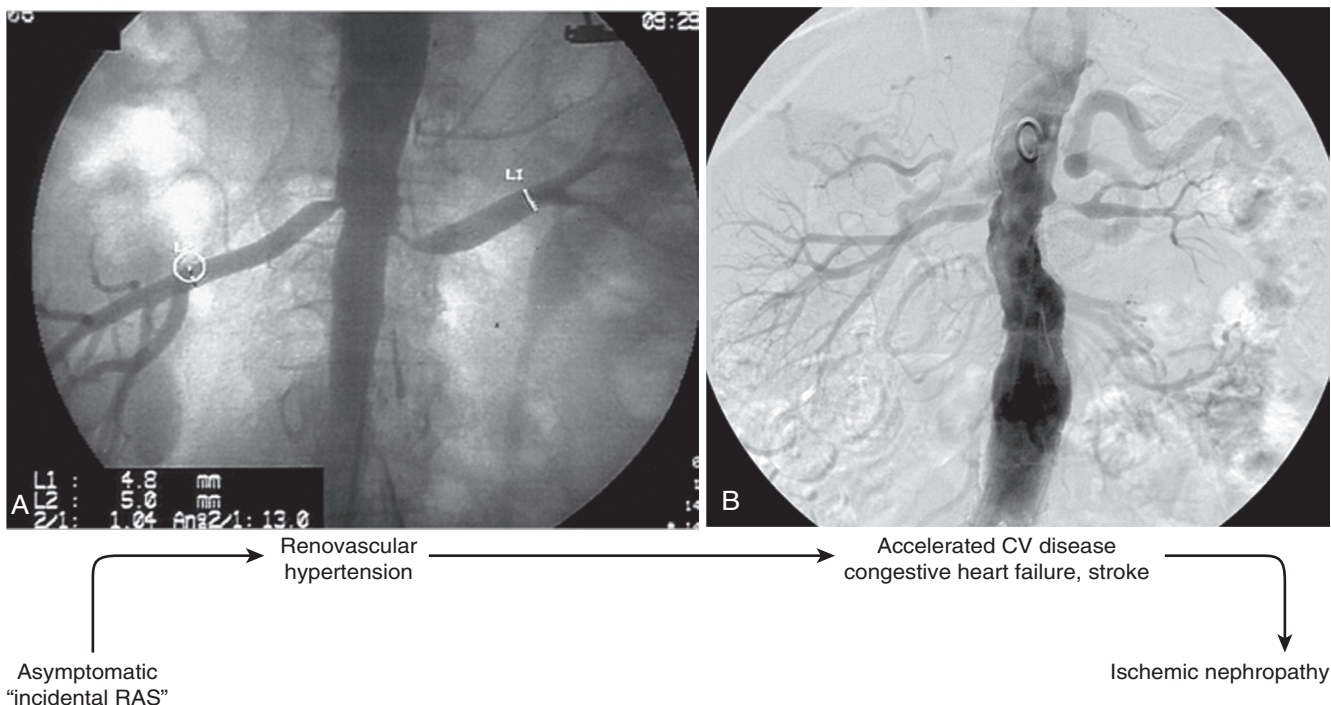
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The evaluation and treatment of renovascular disease (RVD) overlaps with numerous medical disciplines and subspecialties including nephrology, internal medicine, cardiovascular diseases, interventional radiology, and vascular surgery. These subspecialty groups often deal with different patient subgroups and clinical issues that shape different points of view. Nephrologists, for example, regularly encounter refractory hypertension and/or declining kidney function with high-grade stenosis to a solitary functioning kidney. Cardiologists more commonly manage patients with refractory congestive heart failure at risk for “flash” pulmonary edema than internists who may deal with hypertensive patients with progressive hypertension or a rise in serum creatinine (Fig. 47.1). All these conditions can represent clinical manifestations of RVD but present different comorbid risk and management issues. Not surprisingly, perceptions related to renovascular hypertension and ischemic nephropathy differ even among informed clinicians. Negative results from prospective randomized trials, such as the Angioplasty and Stenting for Renal Artery Lesions (ASTRAL) in the United Kingdom and Cardiovascular Outcomes in Renal Artery Lesions (CORAL) in the United States comparing optimal medical therapy with or without endovascular stent procedures, have been controversial.<sup>1,2</sup> Reports for “high-risk” subsets not enrolled in those studies identify major clinical and

mortality benefits from restoring renal blood flow.<sup>3-5</sup> These diverse observations underscore the ambiguity clinicians encounter in practice. Ultimately, RVD and impaired blood flow affect not only blood pressure and cardiovascular risk but also threaten the viability of the kidney. These can lead to irreversible loss of kidney function, sometimes designated “ischemic nephropathy” or “azotemic RVD.”<sup>6,7</sup> It is therefore important that nephrologists have a solid foundation related to the implications of reduced renal perfusion and the risks and benefits of both medical management and restoration of renal artery patency.

## HISTORICAL PERSPECTIVE

RVD is among the most extensively studied forms of secondary hypertension. Early observations regarding blood pressure regulation revealed important connections among fluid volume, renal arterial pressure, and vascular resistance. The sequence of these observations related to identification of the renin-angiotensin-aldosterone system has been reviewed by Basso and Terrano.<sup>8</sup> Experimental models of 2-kidney and 1-kidney “Goldblatt” hypertension (animals undergo clipping of one renal artery to reduce blood flow, with or without nephrectomy of the contralateral kidney, i.e., 2-kidney-1-clip and 1-kidney-1-clip models)



**Fig. 47.1 Spectrum of atherosclerotic RVD (ARVD).** Panel A depicts an aortogram obtained during coronary angiography demonstrating moderate “incidental” stenosis of both renal arteries in a 67-year-old man with symptomatic coronary disease. Panel B illustrates more severe occlusive disease observed in a 68-year-old woman presenting with severe hypertension and episodes of “flash pulmonary edema.” RVD commonly develops in the setting of atherosclerotic disease elsewhere and may be associated with multiple clinical syndromes ranging from renovascular hypertension to accelerated cardiovascular decompensation and ischemic nephropathy. RAS, Renal artery stenosis. (Modified from Herrmann SM, Saad A, Textor SC. Management of atherosclerotic renovascular disease after Cardiovascular Outcomes in Renal Atherosclerotic Lesions (CORAL). *Nephrol Dial Transplant*. 2015;30(3):366–375 with permission.)

represent some of the most widely applied models of blood pressure and cardiovascular regulation. Some of the major milestones in identification and treatment of renovascular hypertension are summarized in [eFig. 47.1](#).

Establishing the relationship between the kidney and blood pressure control in RVD led to efforts to restore renal blood flow with vascular repair, either surgically or using endovascular procedures. Developments in the 1980s and 1990s led to both improved medications and the expanded utilization of percutaneous angioplasty and stents.<sup>9</sup> These both broadened the options for treating patients with vascular disease and raised new issues regarding timing and overall goals of intervention. Further developments highlighted the need for intensive cardiovascular risk factor reduction and more stringent standards of blood pressure control. Broad application of angiotensin-converting enzyme inhibitors and angiotensin receptor antagonists, for reasons other than hypertension alone, changed the clinical presentation of disorders associated with renovascular disease. The challenge for clinicians is how and when to most effectively apply these tools in the management of individual patients.<sup>6,10</sup>

## **PATHOPHYSIOLOGY OF RENOVASCULAR HYPERTENSION AND ISCHEMIC NEPHROPATHY**

### **“RENAL ARTERY STENOSIS” VERSUS “RENOVASCULAR HYPERTENSION”**

The presence of a renovascular abnormality alone does not translate directly into functional importance. Some degree of RVD can be identified in up to 20% to 45% of patients undergoing vascular imaging for other reasons, particularly coronary angiography or lower extremity peripheral vascular disease.<sup>11,12</sup> Most “incidentally detected” stenoses are of minor hemodynamic and/or clinical significance. The term “renovascular hypertension” refers to a rise in arterial pressure induced by reduced renal perfusion. A variety of lesions can lead to the syndrome of renovascular hypertension, some of which are listed in [Box 47.1](#). Studies of vascular obstruction using latex rubber casts indicate that between 70% to 80% of lumen obstruction must occur before changes in blood flow or pressure across the lesion can be detected.<sup>13</sup> Measurements of pressure gradients during renal angiography confirm that a pressure gradient of at least 10 to 20 mm Hg between the aorta and poststenotic renal artery is required before measurable changes in renin release occur.<sup>14,15</sup> When advanced stenosis is present, the pressure and flow drop steeply as illustrated in [Fig. 47.2](#). When lesions have reached this degree of hemodynamic significance, they are deemed “critical” stenoses.

Upon developing critical stenosis, a series of events leads to a rise in systemic arterial pressure to restore renal perfusion pressure, as illustrated in [Fig. 47.3](#). One can view development of rising pressures in this context as an integrated renal response to maintain renal perfusion of the poststenotic kidney. It is important to distinguish between experimental models of “clip” stenosis, at which time a sudden change in renal perfusion is induced, and the more common clinical situation of gradually

### **Box 47.1 Examples of vascular lesions producing renal hypoperfusion and the syndrome of renovascular hypertension**

#### **Unilateral disease (analogous to 1-clip-2-kidney hypertension)**

- Unilateral atherosclerotic renal artery stenosis
- Unilateral fibromuscular dysplasia (FMD)<sup>200</sup>
  - Medial fibroplasia
  - Perimedial fibroplasia
  - Intimal fibroplasia
  - Medial hyperplasia
- Renal artery aneurysm
- Arterial embolus
- Arteriovenous fistula (congenital/traumatic)
- Segmental arterial occlusion/dissection (posttraumatic)
- Extrinsic compression of renal artery (e.g., pheochromocytoma)
- Renal compression (e.g., metastatic tumor)

#### **Bilateral disease or solitary functioning kidney (analogous to 1-clip-1-kidney model)**

- Stenosis to a solitary functioning kidney
- Bilateral renal arterial stenosis
- Aortic coarctation/mid-aortic syndrome (sometimes seen in children)
- Systemic vasculitis (e.g., Takayasu and polyarteritis)
- Atheroembolic disease
- Vascular occlusion due to endovascular aortic stent graft

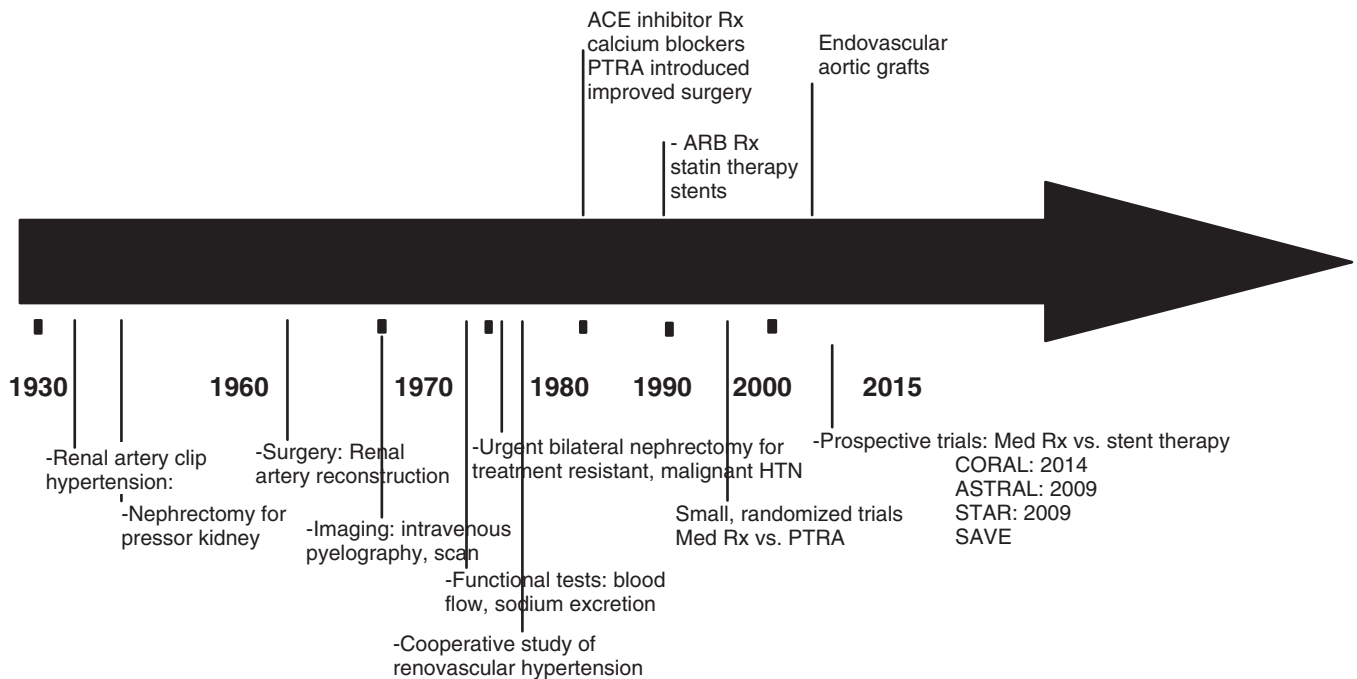
progressive lumen obstruction. In the latter instance, hemodynamic characteristics change slowly and are likely to produce hypertension over a prolonged time interval. The rise in systemic pressure restores normal renal perfusion, often with normal-sized kidneys and no discernible hemodynamic compromise. If the renal artery lesion progresses further (or is experimentally advanced), the cycle of reduced perfusion and rising arterial pressures repeats until malignant-phase hypertension develops.<sup>16</sup>

A corollary to “critical” arterial stenosis is that treatment aimed at normalization of elevated systemic pressures in renovascular hypertension reduces renal perfusion pressure beyond the stenotic lesion. Poststenotic pressures may fall to levels where autoregulation can no longer maintain blood flow. This underperfusion of the kidney activates counter-regulatory pathways and leads to a sequence of events again directed toward restoring kidney perfusion.

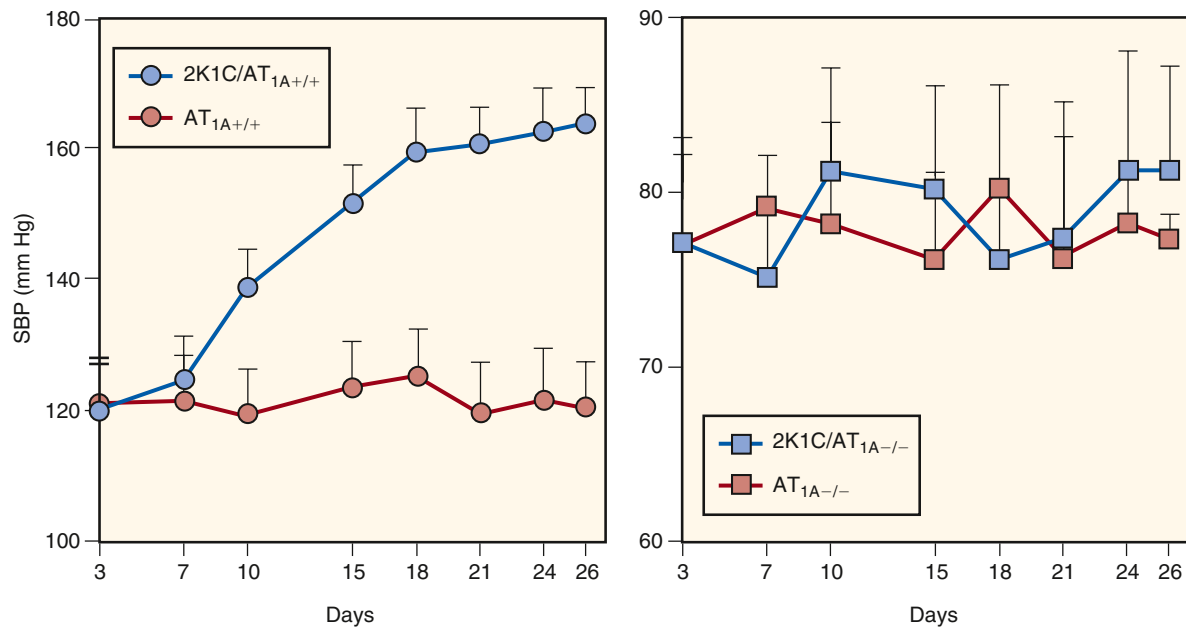
### **THE ROLE OF THE RENIN-ANGIOTENSIN SYSTEM IN 1-KIDNEY AND 2-KIDNEY RENOVASCULAR HYPERTENSION**

Reduction in renal perfusion pressures activates release of renin from juxtaglomerular cells within the affected kidneys. Animals genetically modified to lack the angiotensin 1 (AT1) receptor fail to develop 2-kidney-1 clip hypertension. This is illustrated in [eFig. 47.2](#).<sup>17</sup> Experiments using kidney transplantation from AT1 receptor knockout mice indicate that both systemic and renal angiotensin receptors participate in blood pressure regulation in additive fashion.<sup>18</sup>

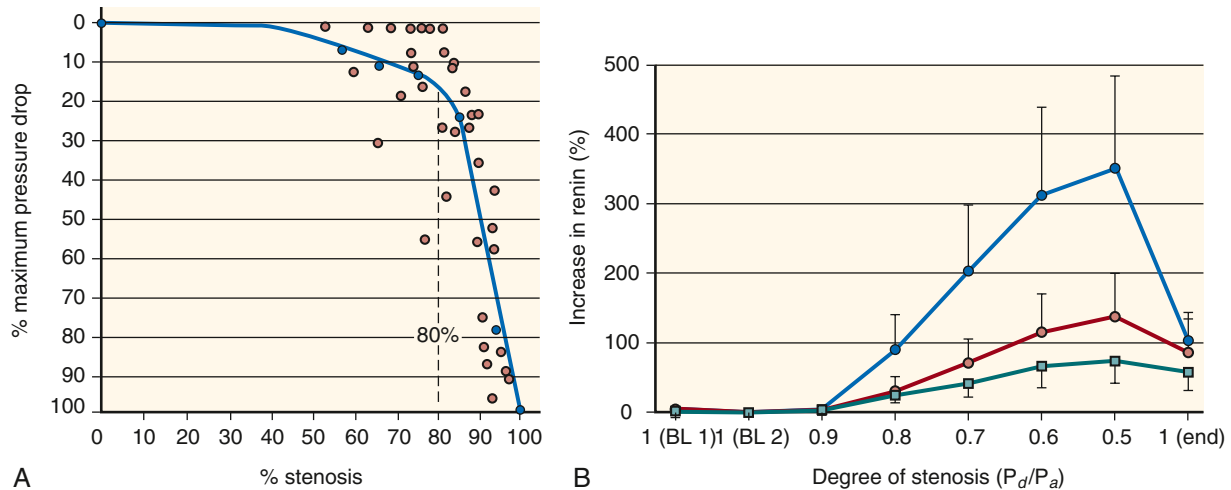
### Timeline of milestones related to renovascular hypertension



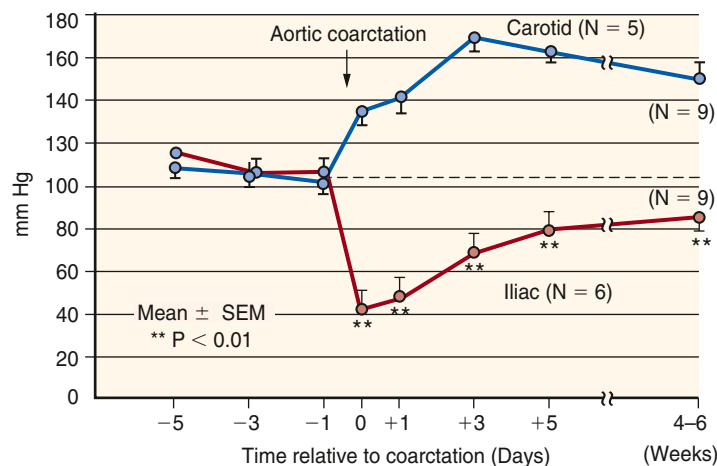
**eFig. 47.1 Historical timeline of major milestones in renovascular disease (RVD).** Although pressor substances derived from the kidney were identified more than a century ago, recognition that renal perfusion regulates arterial blood pressure occurred only in the 1930s. Technical advances allowing surgical reconstruction, more effective arterial imaging, and endovascular revascularization gradually evolved from 1960 to the present era. Important advances in antihypertensive drug therapy, particularly with agents that block the renin-angiotensin system, and other advances in managing atherosclerotic disease such as statins, continue to define optimal medical management of patients with RVD. The introduction of endovascular aortic stent grafts represents an iatrogenic form of RVD that benefits from protection of the renal artery patency. Results from several prospective, randomized clinical trials indicate that renal revascularization often fails to add substantial benefits to effective medical therapy in the near term for patients with moderate atherosclerotic RVD, although high-risk subsets are an exception. *ACE*, Angiotensin-converting enzyme; *ARB*, angiotensin receptor blocker; *PTRA*, percutaneous transluminal renal angioplasty.



**eFig. 47.2 Systolic blood pressures in mice before and after placement of a renal artery clip in experimental two-kidney, one-clip renovascular hypertension (2K1C).** The rise in systolic blood pressure (SBP) after clip placement develops rapidly only in mice with an intact angiotensin 1A receptor (AT<sub>1A</sub><sup>+/+</sup>; blue circles, left panel). This rise is blocked by administration of an angiotensin receptor blocker (red circles, left panel). A genetic knockout mouse strain with no AT1A receptor (AT<sub>1A</sub><sup>-/-</sup>) has a lower SBP and no change after renal artery clipping (blue squares, right panel). No additional effect is noted with an angiotensin receptor blocker (red squares, right panel). These data reinforce the essential role of the renin-angiotensin system and an intact angiotensin 1 receptor for the development of renovascular hypertension. (Modified from Cervenka L, Horacek V, Vaneckova I, et al. Essential role of AT1-A receptor in the development of 2K1C hypertension. *Hypertension*. 2002;40:735–741.)



**Fig. 47.2** (A) Measured fall in arterial pressure and blood flow across stenotic lesions induced in experimental animals. The degree of stenosis was determined using latex casts after completion of the experiment. These data indicate that “critical” lesions require 70% to 80% luminal obstruction before hemodynamic effects can be detected. (B) Effects of a balloon-induced, unilateral, controlled, graded stenosis (expressed as  $P_d/P_a$  ratio) on plasma renin concentration in the aorta (green), in the vein of the stenotic kidney (blue), and in the vein of the non-stenotic kidney (red).  $P_a$ , Aortic pressure;  $P_d$ , distal pressure. (A from May AG, Van de Berg L, DeWeese JA, et al. Critical arterial stenosis. *Surgery*. 1963;54:250–259; B from De Bruyne B, Manoharan G, Pijls NHJ, et al. Assessment of renal artery stenosis severity by pressure gradient measurements. *J Am Coll Cardiol*. 2006;48:1851–1855.)

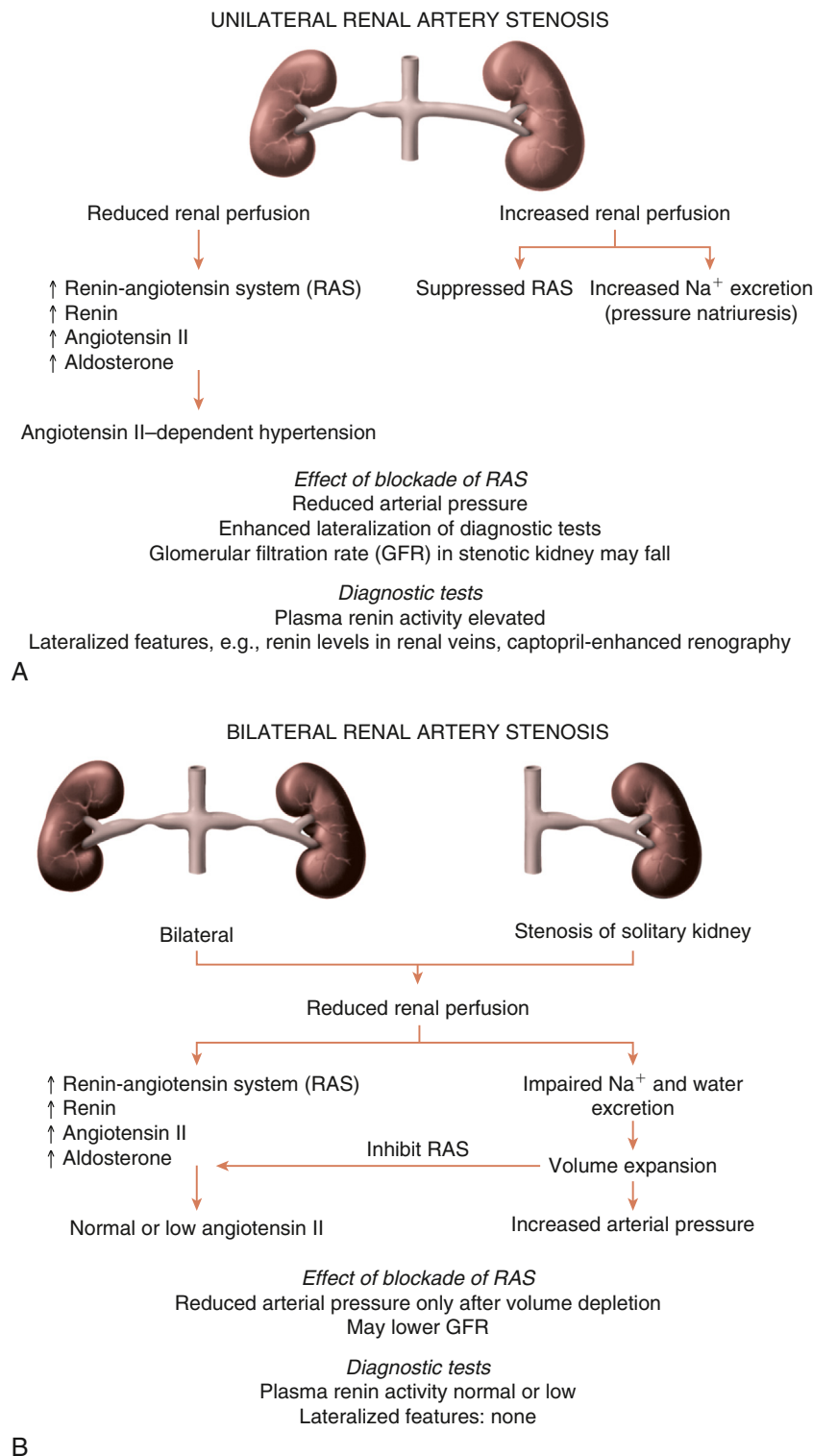


**Fig. 47.3** Systemic arterial pressure (carotid) and poststenotic renal perfusion pressures (iliac) in an aortic coarctation model with a clip placed between the right and left renal arteries. Measurements were obtained in conscious animals during development of renovascular hypertension. They illustrate the fact that despite a persistent gradient across the stenosis, renal perfusion pressure (inferred from iliac pressure) returns to near normal levels as a result of systemic hypertension. A corollary of this pressure gradient is that reduction of systemic pressures lowers poststenotic perfusion pressures, sometimes below the range of autoregulation. (From Textor SC, Smith-Powell L. Post-stenotic arterial pressures, renal haemodynamics and sodium excretion during graded pressure reduction in conscious rats with one- and two-kidney coarctation hypertension. *J Hypertens*. 1988;6(4):311–319 with permission.)

The role of the renin-angiotensin axis in renovascular hypertension depends, in part, on whether or not a contralateral, nonstenotic kidney is present. Most human renovascular hypertension is considered analogous to 2-kidney-1-clip experimental (“Goldblatt”) hypertension (a clip is applied to one renal artery in the rat to cause stenosis, and the other renal artery is untouched). The contralateral, nonstenotic kidney is subjected to elevated systemic perfusion pressures. Rising perfusion pressure forces natriuresis from the nonstenotic kidney and suppresses renin release. Hence the nonstenotic kidney tends to prevent the rise in systemic pressures, thereby perpetuating reduced perfusion to the stenotic side and promoting continued renin release from the stenotic kidney. Hypertension in these models is angiotensin dependent and is associated with elevated circulating levels of plasma renin activity, as illustrated in Fig. 47.4. The 2-kidney-1-clip model of renovascular hypertension provided the basis for many of the early functional studies of surgically curable hypertension in which side-to-side function was compared

regarding glomerular filtration, sodium excretion, etc. It also forms the basis for comparing kidneys side to side using radionuclide studies, such as captopril renograms, and renal vein renin determinations.

When no such contralateral kidney is present or able to respond to pressure natriuresis, the mechanisms sustaining hypertension differ. This corresponds to the 1-kidney-1-clip hypertensive model (the rat undergoes unilateral nephrectomy and clipping of the contralateral renal artery) (Fig. 47.4B). Although renin release occurs initially, elevated systemic pressures develop with sodium and volume retention due to ineffective natriuresis. Rising pressures eventually restore renin levels to normal. Hypertension in this model is not demonstrably dependent on angiotensin II unless prior sodium depletion is achieved. Clinical examples of this situation include bilateral renal artery stenosis (RAS) or stenosis in a solitary functioning kidney where the entire renal mass is affected, such as occurs with orthotopic renal transplants. In such cases, diagnostic comparison of side-to-side renin release is not possible or has little meaning.

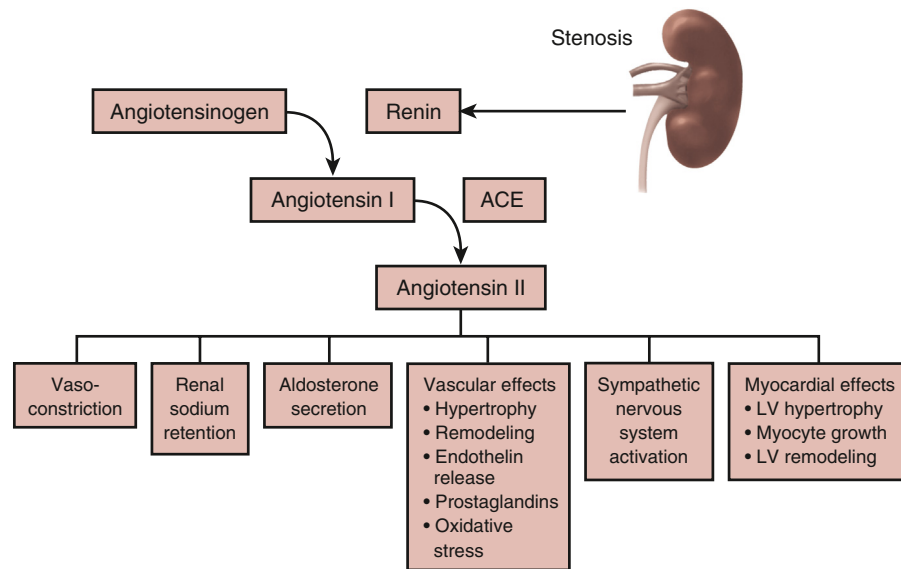


**Fig. 47.4 Schematic view of two-kidney (A) and one-kidney (B) renovascular hypertension.** These models differ by the presence of a contralateral kidney exposed to elevated perfusion pressures in two-kidney hypertension. The nonstenotic kidney tends to allow pressure natriuresis to ensue and produces ongoing stimulation of renin release from the stenotic kidney. The one-kidney model eventually produces sodium retention and a fall in renin with minimal evidence of angiotensin dependence unless sodium depletion is achieved.

### MECHANISMS SUSTAINING RENOVASCULAR HYPERTENSION

Recruitment of numerous additional pressor pathways increases the complexity of managing hypertension in RVD. Identification of components of the renin-angiotensin

system provided a crucial link to understanding several of these systems. Circulating renin is derived primarily from the kidney in response to a reduction in renal perfusion pressure detected by loss of afferent arteriolar stretch.<sup>19</sup> Renin itself has biological activity directed mainly at



**Fig. 47.5 Schematic view of activation of the renin-angiotensin system beyond a renal artery stenotic lesion.** Generation of circulating and local angiotensin II leads to widespread effects including sodium retention, efferent arteriolar vasoconstriction, and elevated systemic vascular resistance. Additional studies implicate angiotensin II in many other pathways of vascular and cardiac smooth muscle remodeling, activation of inflammatory and fibrogenic cytokines, coagulation factors, and induction of other vasoactive systems. ACE, Angiotensin-converting enzyme.

the enzymatic release of angiotensin I (AngI) from its circulating substrate, angiotensinogen, in plasma and possibly other sites.<sup>8</sup> Hence the signal of reduced kidney pressures is amplified and transmitted through a major systemic vasopressor system and is one mechanism by which renovascular hypertension develops.

### RENIN-ANGIOTENSIN SYSTEM

The renin-angiotensin system has effects beyond its vasoconstrictor action in renovascular hypertension,<sup>20,21</sup> as illustrated in Fig. 47.5. Activation of this system increases vascular resistance, sodium retention, and aldosterone stimulation. Complex interactions between angiotensin II and tissue and cellular systems occur, leading to vascular remodeling, left-ventricular hypertrophy, and activation of inflammatory and fibrogenic mechanisms.

RVD also leads to disturbances in sympathetic nerve signaling, which may differ between 1-kidney and 2-kidney models. Muscle sympathetic nerve activity is increased in humans with renovascular hypertension, and blood pressure responses to adrenergic inhibition are amplified.<sup>22</sup>

A major transition occurs with altered oxidative stress within the systemic vasculature, leading to increased oxygen free radicals.<sup>23</sup> Experimental animals with 2-kidney-1-clip hypertension develop an increase in oxidative stress that can be reversed, in part, with angiotensin blockade and/or antioxidants.<sup>24</sup> Some of these changes are reversible after revascularization in patients with both atherosclerotic and fibromuscular RVD.<sup>25</sup> Studies in a swine model demonstrate an interaction between “atherosclerosis” induced by cholesterol feeding and RVD.<sup>26,27</sup> These data are supported by observations of a rise in nitric oxide (NO) and reduction in malondialdehyde within 24 hours of endovascular revascularization in atherosclerotic RVD.<sup>28</sup> In some studies, oxidative stress can be reversed by both infusion of antioxidants and successful revascularization.<sup>20</sup> Experimental infusion of agents that

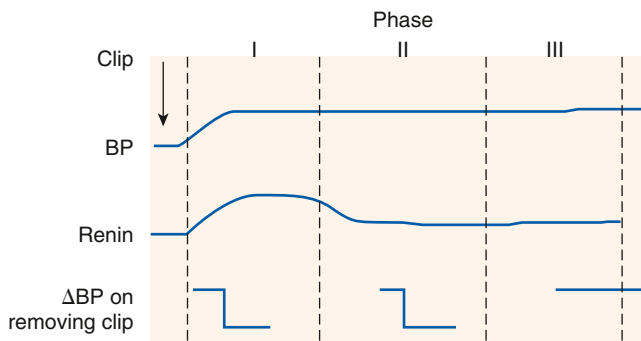
protect mitochondria from reactive oxygen species, by inhibiting opening of the mitochondrial transition pore, is associated with improved recovery of microvascular structures and renal function after revascularization.<sup>29</sup>

### PHASES OF DEVELOPMENT OF RENOVASCULAR HYPERTENSION

Experimental models of renovascular hypertension indicate that mechanisms sustaining hypertension change over time (eFig. 47.3).<sup>30</sup> Even before occlusive lesions are evident, accumulation of inflammatory monocytes in the renal vascular wall and elevation of circulating cytokines are evident.<sup>31</sup> An early pressor phase is characterized by elevated circulating renin activity and hypertension, both of which return to normal after removing the vascular lesion (e.g., the clip). A second phase is characterized by a return of circulating renin activity to normal or low levels, during which hypertension persists and blood pressure can still respond to clip removal. A third phase occurs during which removal of the clip no longer leads to reduction in arterial pressure. These observations highlight the transition between differing mechanisms of vascular resistance, some of which become independent of renal perfusion. Whether these phases translate directly to human RVD is not well known, but they likely affect the period of time within which renal revascularization can reverse this process, as illustrated in Fig. 47.6.

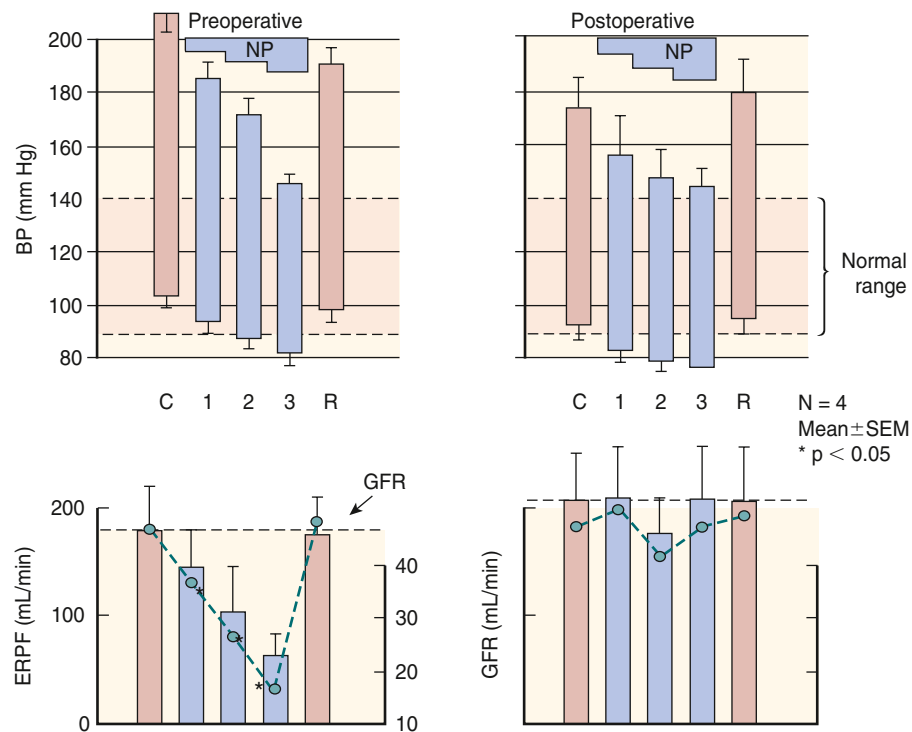
### MECHANISMS OF “ISCHEMIC NEPHROPATHY”

Reduced renal perfusion beyond “critical” stenosis ultimately leads to loss of viable kidney function, as illustrated in eFig. 47.4.<sup>32</sup> Patients with stenosis affecting the entire renal mass develop reduced blood flow and reduced glomerular filtration when poststenotic pressures fall below the range of autoregulation. This process can be reversible if pressure is restored and/or the vascular lesion is removed. The

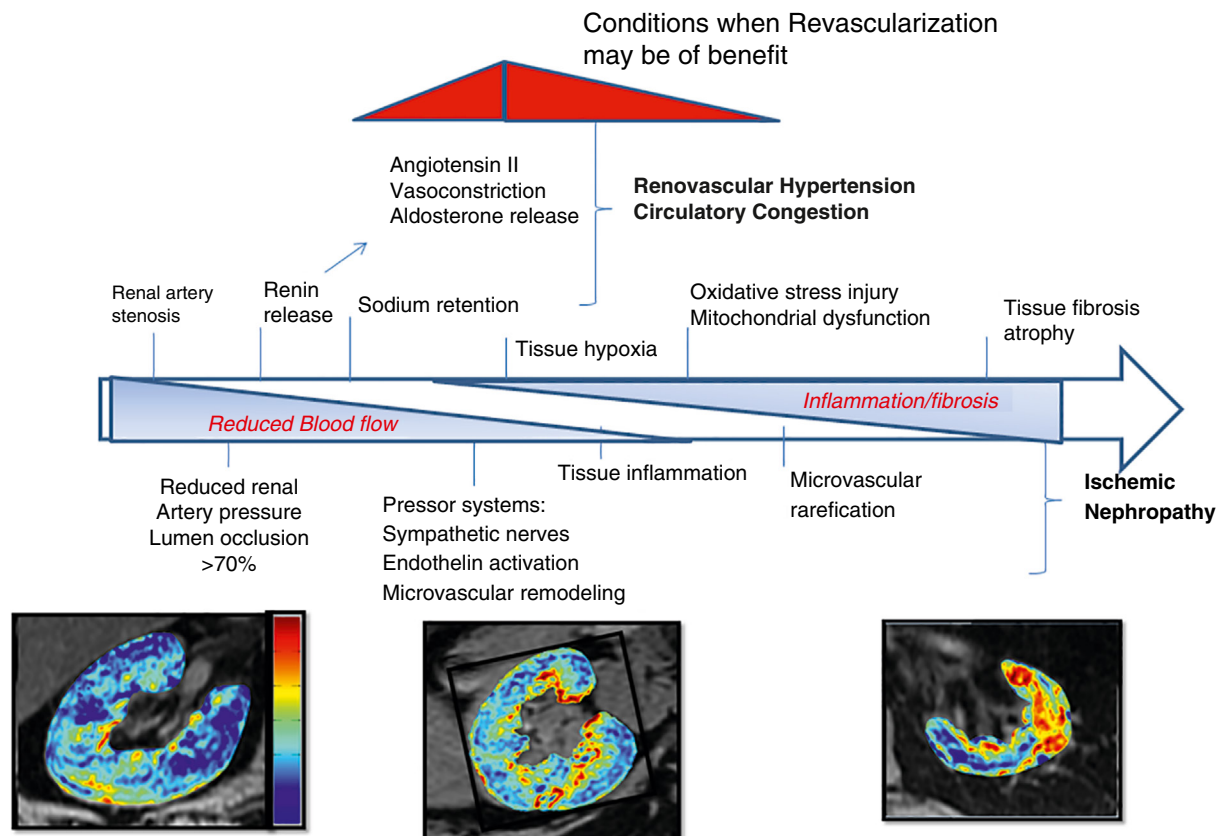


**eFig. 47.3 Schematic depiction of phases observed in experimental renovascular hypertension.** Initially, high levels of renin activity fall in the chronic phase, although removal of the renal artery clip corrects hypertension. These observations support the concept of renal artery stenosis leading to the recruitment of additional structural and pressor mechanisms after initial activation of the renin-angiotensin system. The degree to which human renovascular hypertension follows these patterns is not well known. *BP*, Blood pressure. (From Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al., eds. *Brenner and Rector's: The Kidney*. 10th ed. Philadelphia: Elsevier; 2016:1567–1609.)

#### BLOOD PRESSURE, RENAL PLASMA FLOW, AND GFR IN PATIENTS WITH BILATERAL RAS DURING NITROPRUSSIDE INFUSION BEFORE AND AFTER REVASCULARIZATION



**eFig. 47.4 Effective renal plasma flow (ERPF) and glomerular filtration rate (GFR) in patients with critical bilateral renal artery stenosis during pressure reduction with sodium nitroprusside (NP).** Reducing systemic blood pressure to normal levels produced a reversible fall in both plasma flow and GFR. Repeat studies in the same patients (right-hand panel) after unilateral surgical revascularization demonstrate that the sensitivity of blood flow and GFR to pressure reduction can be reversed. *SEM*, Standard error of the mean. (From Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al., eds. *Brenner and Rector's: The Kidney*. 10th ed. Philadelphia: Elsevier; 2016:1567–1609.)



**Fig. 47.6 Progression of renovascular disease and the limitations of revascularization.** Central horizontal arrow depicts the sequence of events associated with progressive renovascular occlusion. Reductions in blood flow and perfusion pressure develop only after substantial (>70%) lumen occlusion, leading to activation of the renin-angiotensin system. Bottom images depict blood-oxygen-level-dependent (BOLD) MR slices mapping deoxyhemoglobin within the kidney. Despite reduced blood flow, renal oxygenation is significantly reduced (bottom right panel) only with severe and prolonged occlusion, which is associated with activation of inflammatory injury, vascular rarefaction, and tissue fibrosis. Renal revascularization can restore kidney perfusion and reverse these processes only under conditions that have not become irreversible (depicted in red). (Reprinted with permission from Bhalla V, Textor SC, Beckman JA, et al. Revascularization for renovascular disease: a scientific statement from the American Heart Association. *Hypertension*. 2022;79(8):e128–e143. © 2022 American Heart Association, Inc.)

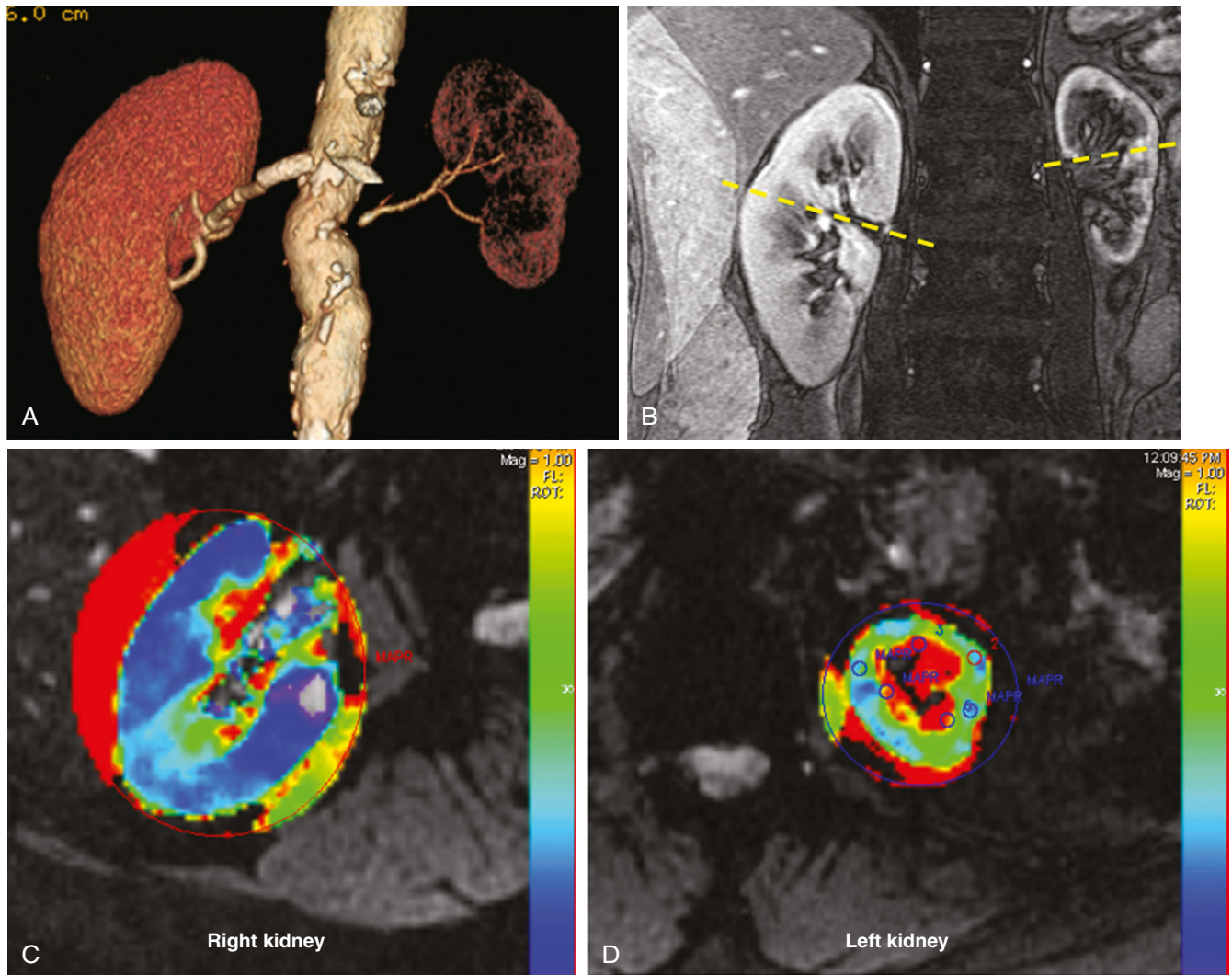
mechanisms by which this occurs differ from those that govern the development of hypertension.<sup>15,33</sup>

### ADAPTIVE MECHANISMS TO REDUCED RENAL PERFUSION

The kidney, brain, and heart are all highly perfused with oxygenated blood. Yet unlike brain or cardiac tissue, oxygen extraction by the kidney is much lower, consistent with its function as a filtering organ. Under basal conditions, renal blood flow is among the highest of all organs reflecting its filtration function. Less than 10% of delivered oxygen is sufficient to maintain overall renal metabolic needs. The oxygen consumption that does occur within the kidney reflects solute transport. Measurements of both renal vein oxygen saturation and erythropoietin in patients with high-grade renovascular lesions indicate that whole organ “ischemia” is rarely present.<sup>34</sup> The kidney maintains autoregulation of blood flow in the face of reduced arterial diameters of up to 75%. Imaging using blood-oxygen-level-dependent magnetic resonance (BOLD MR) demonstrates reduced oxygenation in deeper medullary regions in both normal kidneys and patients with RVD despite preserved cortical oxygenation.<sup>33</sup> The renal medulla normally functions at levels close to hypoxia and is therefore sensitive

to acute changes in perfusion.<sup>35</sup> Studies of patients with RVD sufficient to reduce blood flow, kidney volume, and GFR indicate remarkable preservation of tissue oxygenation using BOLD MR.<sup>36</sup> This stability of oxygen gradients is due partly to the surplus of oxygenated blood perfusing the cortex and to reduced oxygen consumption as a result of reduced solute filtration leading to reduced reabsorptive work of the affected kidney.<sup>37</sup> When sufficiently severe, however, reduced blood flow leads to accumulation of deoxygenated molecular hemoglobin (Fig. 47.7).<sup>36,38</sup>

More severe vascular occlusion produces both cortical and expanded medullary hypoxia as these adaptive measures are overwhelmed and tissue injury ensues.<sup>37</sup> Eventually, structural atrophy of the renal tubules occurs, partly due to necrosis and apoptosis.<sup>39</sup> The latter is an active, programmed cellular death that appears to be closely regulated and differs from tissue necrosis. Tubular atrophy is potentially reversible, and the kidney maintains the capacity for tubular cell regeneration under many conditions. These features support the concept that underperfused kidney tissue sometimes can achieve a “hibernating” state capable of restoring function if blood flow is restored.<sup>40</sup> In late stages, histologic examination demonstrates reduced glomerular volume, loss of tubular structures near underperfused glomeruli, and areas of local inflammation as illustrated in eFig. 47.5A and B.<sup>41</sup>



**Fig. 47.7** Computed tomography angiogram (A) of high-grade vascular occlusion leading to left kidney atrophy while the right kidney is well perfused beyond an endovascular stent. T2 signals from magnetic resonance imaging (B) illustrate a gradient of perfusion from cortex to lower levels in the deeper medullary sections, especially in the left kidney. Mapping of the blood-oxygen-level-dependent (BOLD) magnetic resonance  $R2^*$  levels (a function of deoxyhemoglobin) (C and D) illustrates the normal gradient in the right kidney from cortex (low  $R2^*$ , blue) to deeper medullary segments that have progressively higher levels of  $R2^*$  (green to red). The  $R2^*$  map of the left kidney (D) illustrates overt cortical hypoxia evident beyond critical vascular occlusion associated with a larger fraction of the slice having high  $R2^*$  values (red) associated with fractional medullary hypoxia.

A reduction in renal perfusion leads to diminished “shear stress” distal to the stenosis.<sup>42</sup> This condition reduces production of NO and accelerates release of renin and generation of angiotensin II in the stenotic kidney. A direct consequence of reduced perfusion within the poststenotic kidney is progressive rarefaction of small vessels in both cortex and medulla, eventually producing tubular collapse<sup>43</sup> (eFig. 47.6). Glomerulosclerosis is a late event and usually reflects severe loss of glomerular filtration rate. Experimental studies infusing autologous endothelial progenitor cells and/or vascular endothelial growth factors (VEGF) into the renal artery demonstrate that functional recovery of GFR and restoration of vascular function can sometimes be achieved, suggesting that angiogenesis may be an achievable goal in this vascular bed.<sup>44,45</sup> Taken together, the kidney is subject to a wide variety of vasoactive and inflammatory mediators, which can be disturbed by loss of blood flow and perfusion pressure. These disturbances appear to activate a variety of fibrogenic and local destructive mechanisms,

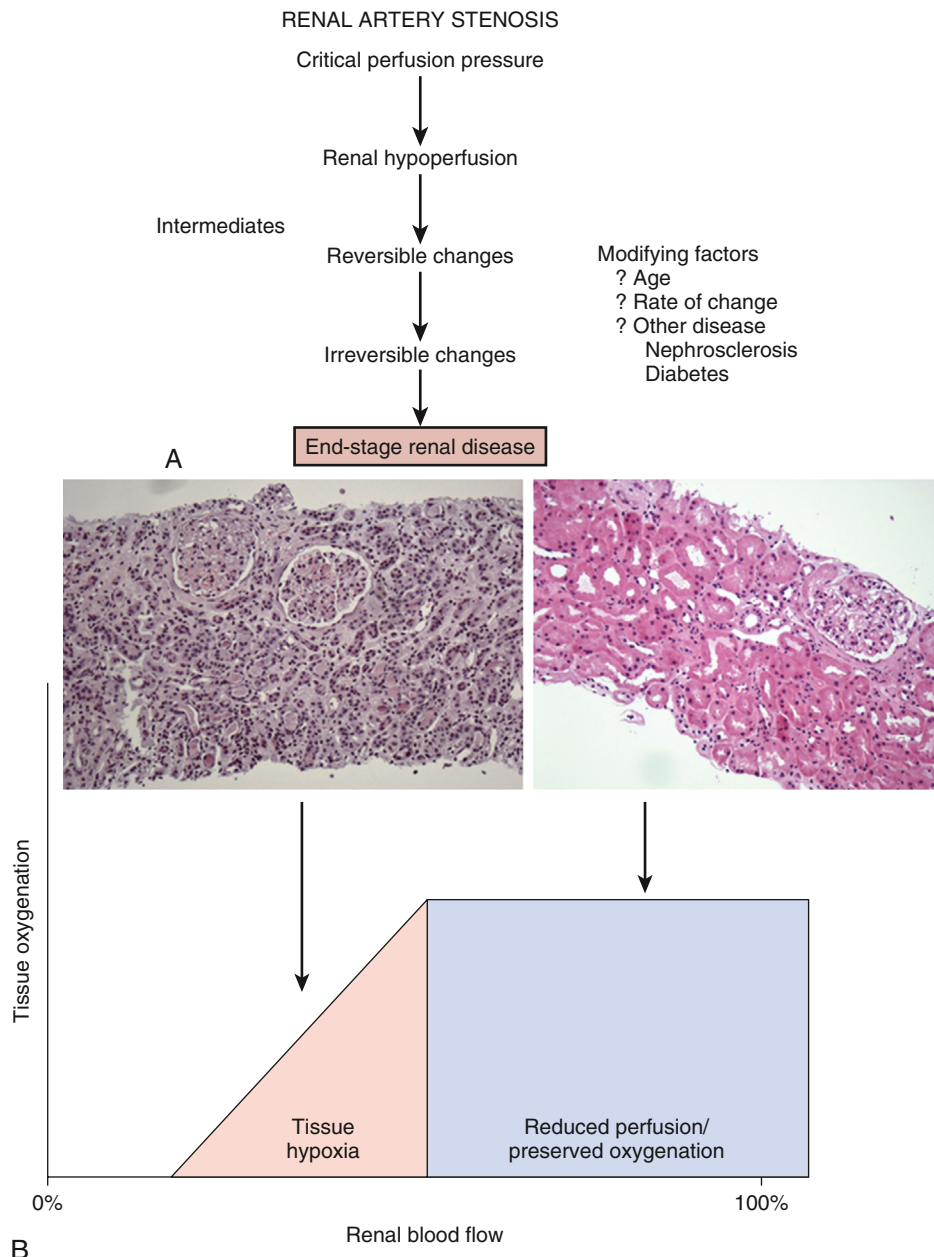
which can lead to irreversible parenchymal damage within the kidney. For further discussion of vasoactive and inflammatory molecules in the kidney, see Chapter 11.

### CONSEQUENCES OF RESTORING RENAL BLOOD FLOW

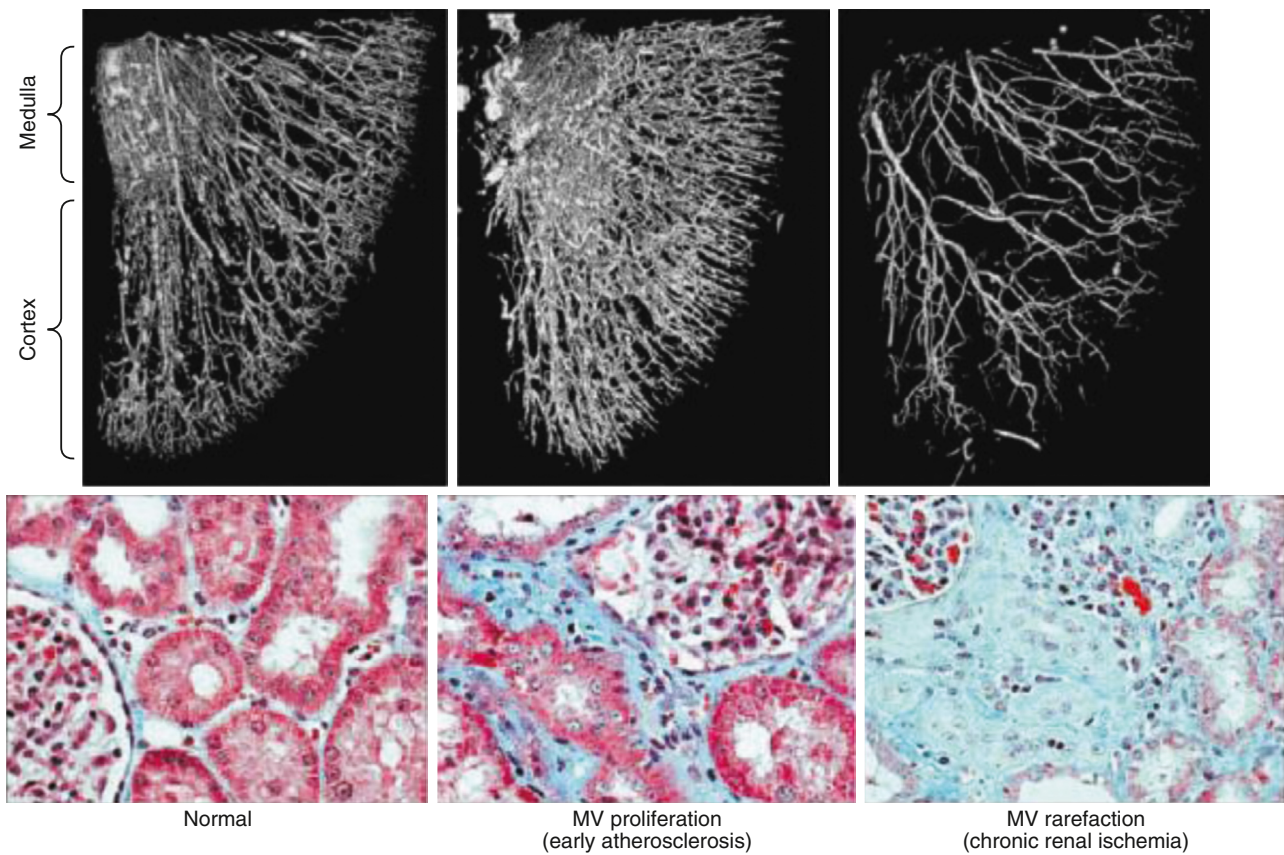
As illustrated in eFig. 47.6, restoring renal perfusion can allow recovery of renal function if accomplished when these changes remain reversible. At some point, both inflammatory and fibrogenic mechanisms appear no longer to respond with recovery of renal function.

### RENAL REPERFUSION INJURY

The course of recovery after restoration of blood flow in an underperfused kidney depends on the extent and duration of the reperfusion injury, in addition to the adequacy of reperfusion.<sup>46</sup> Paradoxically, some tissues subjected to ischemia



**eFig. 47.5** (A) General clinical paradigm that a reduction in the renal blood flow related to hypoperfusion of the kidney can be reversible but can undergo a transition to irreversible injury eventually that no longer responds to restoration of blood flow alone. (B) Schematic depiction of the adaptation of tissue oxygenation in the kidney to moderate reductions in blood flow. Lowering renal blood flow by up to 30% to 40% in human subjects (right side) is associated with preservation of both oxygen gradients (lower graph) and tissue histology (upper images). The fact that such adaptation occurs partly explains the stability of kidney function during antihypertensive drug therapy that lowers systemic pressure and renal blood flow in patients with atherosclerotic renal artery stenosis. Eventually, more severe reductions of blood flow overwhelm “adaptation,” leading to overt tissue hypoxia (left side). (Magnification: low power:  $\times 40$ .) (Modified from Gloviczki ML, Keddiss MT, Garovic VD, et al. TGF expression and macrophage accumulation in atherosclerotic renal artery stenosis. *Clin J Am Soc Nephrol*. 2013;8(4):546–553; and Textor SC, Lerman LO. Paradigm shifts in atherosclerotic renovascular disease: where are we now? *J Am Soc Nephrol*. 2015;26:2074–2080.)



**eFig. 47.6** *Upper panels*, Microcomputed tomography reconstructions of vessels in experimental atherosclerosis and superimposed large-vessel renovascular disease at an early stage (middle) and after chronic ischemia (right), compared with normal (left). *Lower panels*, Corresponding renal histology (trichrome stain  $\times 400$ ). Complex microvascular dysfunction and rarefaction develop in poststenotic kidneys. These are accelerated and/or modified by angiogenic stimuli, oxidative stress pathways, and a variety of cytokines, leading eventually to interstitial fibrosis. Studies in experimental animals suggest that angiogenic stimuli such as intrarenal infusion of endothelial progenitor cells or mesenchymal stem cells may offer the potential to repair microvascular injury. *MV*, Microvascular. (From Lerman LO, Chade AR. Angiogenesis in the kidney: a new therapeutic target? *Curr Opin Nephrol Hyper*. 2009;18:160–165.)

undergo morphologic and functional changes, which worsen during the reperfusion phase.<sup>47</sup> This is thought to reflect vascular endothelial damage and activation of leukocytes, which may be “primed” to obstruct distal capillaries after restoration of perfusion pressure contributing to a so-called “no-reflow” phenomenon. Under experimental conditions, reperfusion injury appears to require major degrees of pro-oxidant stress with excess PGF2  $\alpha$  isoprostanes and free oxygen radicals, particularly with a deficit of NO.<sup>48</sup> Studies in experimental RVD in swine suggest that pretreatment with a mitochondrial transition pore inhibiting agent allows improved recovery of blood flow, preservation of microvascular integrity, and improved function, which may be consistent with prevention of ischemia/reperfusion injury.<sup>49</sup>

### EPIDEMIOLOGY OF RENAL ARTERY STENOSIS AND RENOVASCULAR HYPERTENSION

The syndrome of renovascular hypertension can be produced by a wide variety of lesions affecting renal blood flow. In unselected mild to moderate hypertensive populations, the frequency appears to be between 0.6% and 3%, whereas in a referral clinic of “treatment-resistant” hypertensive patients, the prevalence may exceed 20%.<sup>50</sup> Some specific example lesions producing renal ischemia are listed in [Box 47.1](#). A rapidly developing form of this disorder can be seen after spontaneous or traumatic renal artery dissection or iatrogenic occlusion, sometimes caused by endovascular aortic stent grafts.<sup>9</sup> The majority of stenotic lesions are either “fibromuscular diseases” or atherosclerotic RVD. The reported prevalence depends heavily on differences between patient groups studied. As noted later, the prevalence of anatomic RVD far exceeds that of renovascular hypertension.

Fibromuscular dysplasia (FMD) commonly refers to one of several noninflammatory conditions affecting the intima or fibrous layers of the vessel wall. In some cases, multiple layers of the vessel wall may be affected. Reports from arteriograms obtained in “normal” renal organ donors indicate that 3% to 5% of individuals may have one of these lesions, many of which do not affect either renal blood flow or arterial pressure.<sup>51</sup> Such lesions can lead to renovascular hypertension, sometimes associated with dissection or progression. Smoking is a risk factor for disease progression. Multifocal lesions identified as “medial fibroplasia” are the most common subtype, often associated with a “string-of-beads” appearance as illustrated in [Fig. 47.8](#).<sup>52,53</sup> These lesions consist primarily of intravascular “webs,” each of which by themselves may have only moderate hemodynamic effect. The combination of multiple webs in series, however, can impede blood flow characteristics and activate responses within the kidney to reduced perfusion. FMD appears in the renal arteries in 65% to 70% of cases and cerebral arteries in 25%.<sup>51</sup> Both renal and cerebral vessels may be abnormal in 10% to 25%. The preponderance of hypertensive cases coming to vascular intervention occurs in women with a bias toward the right renal artery.<sup>54</sup> FMD lesions are classically located away from the origin of the renal artery, often in the midportion of the vessel or at the first arterial bifurcation.<sup>51</sup> Some of these expand to develop vascular aneurysms. Although less common, other dysplastic lesions, particularly intimal hyperplasia, can progress and lead to renal

ischemia and atrophy. Although loss of renal function is unusual with FMD, quantitative imaging of cortical and medullary kidney volumes indicates that parenchymal “thinning” can occur in both the stenotic and contralateral kidneys beyond FMD.<sup>55</sup> FMD accounts for 16% or less of patients referred for renal revascularization for hypertension.<sup>56</sup>

Atherosclerosis affecting the renal arteries is the most common renovascular lesion in the United States. Atherosclerotic renovascular disease can be identified commonly in patients with disease affecting other vascular beds and may be magnified by inflammatory vascular injury.<sup>31</sup> Population-based surveys identify incidental RVD (>60% occlusion by Doppler criteria) in 6.8% of individuals older than age 65 in the United States.<sup>57</sup> A systematic review of imaging studies for other vascular conditions confirms that the prevalence of lesions with more than 50% luminal occlusion rises progressively with the extent of overall atherosclerotic burden.<sup>12</sup> [Table 47.1](#) summarizes multiple reports related to the coexistence of atherosclerotic lesions in various vascular territories. The prevalence of ARVD increases with age and with the presence of atherosclerotic risk factors, such as elevated cholesterol, smoking, and hypertension. The probability of identifying high-grade renal arterial stenosis in hypertensive patients with azotemia rises from 3.2% in the sixth decade to above 25% in the eighth decade.<sup>58</sup>

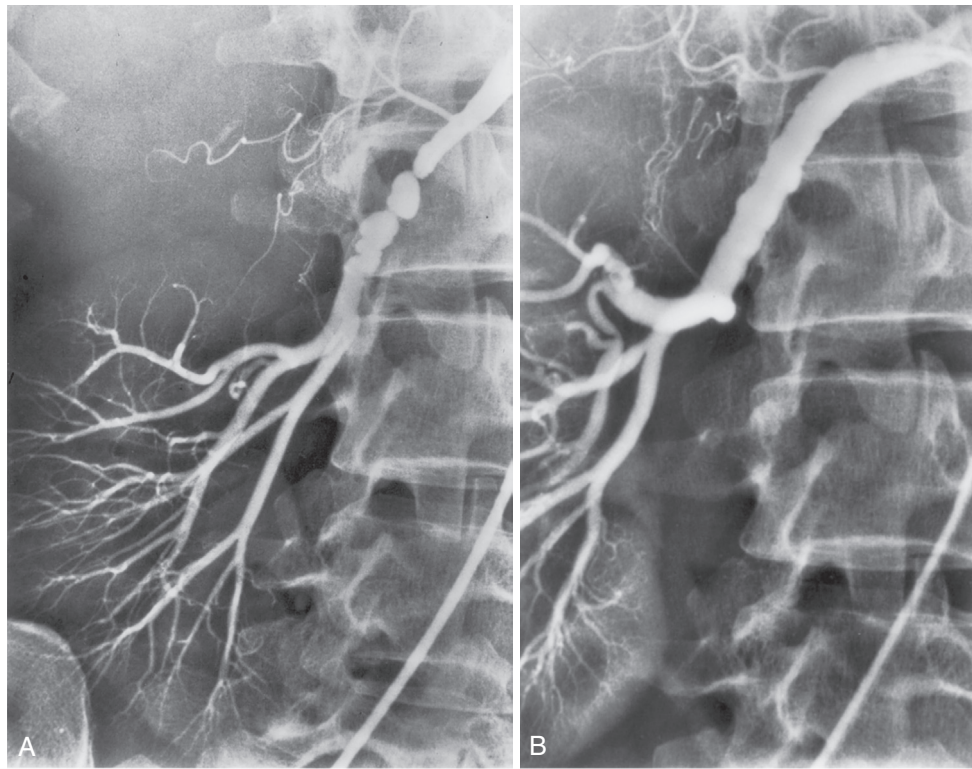
The location of atherosclerotic disease is most often near the origin of the artery, although it can be observed anywhere. Many such lesions represent a direct extension of an aortic plaque into the renal arterial segment. ARVD is strongly associated with preexisting hypertension, cardiovascular lipid risk, diabetes, smoking, and abnormal renal function.<sup>11,59</sup>

### CLINICAL MANIFESTATIONS OF ATHEROSCLEROTIC RENOVASCULAR DISEASE

Manifestations of renal artery disease vary widely across a spectrum illustrated in [Fig. 47.1](#), [Box 47.2](#), and [Table 47.2](#). This spectrum ranges from an incidental finding noted during imaging for other indications to advancing renal failure leading to the need for dialytic support. As lesions progress to more severe levels of occlusion, hypertension develops and multiple manifestations may appear. As described earlier, multiple mechanisms raise systemic arterial pressure and tend to restore renal perfusion pressures to levels close to baseline. [Table 47.2](#) summarizes clinical features associated with ARVD: short duration of hypertension, early age of onset, fundoscopic findings, hypokalemia, etc. Unfortunately, each in isolation has limited discriminatory or predictive value.<sup>60</sup>

As renal artery lesions progress to critical stenosis, they can produce a rapidly developing form of hypertension, which may be severe and associated with polydipsia, hyponatremia, and central nervous system findings.<sup>61</sup> Such cases are most often seen with acute renovascular events, such as sudden occlusion of a renal artery or branch vessel.

More commonly, RVD presents as progressive worsening of preexisting hypertension, often with a modest rise in serum creatinine. Since the prevalence of both hypertension and atherosclerosis increases with age, this disorder must be considered, particularly in older subjects with progressive hypertension. Some of the most striking examples of



**Fig. 47.8 (A and B) Angiographic appearance of medial fibroplasia with serial intravascular webs and small aneurysmal dilatations between them.** These lesions appear in the midportion of the vessel, have a predilection for the right renal artery, and are most common in women. (B) As shown here, these lesions can often be improved substantially by balloon angioplasty.

**Table 47.1 Prevalence rates of atherosclerotic renal artery stenosis<sup>a</sup>**

| Vascular disease                    | Prevalence rate (%)   |
|-------------------------------------|-----------------------|
| Suspected renovascular hypertension | 14.1                  |
| Coronary angiography                | 10.5                  |
| With hypertension                   | 17.8                  |
| Peripheral vascular disease         | 25.3                  |
| AAA                                 | 33.1                  |
| ESKD                                | 40.8 (?) <sup>b</sup> |
| Congestive heart failure            | 54.1 (?) <sup>b</sup> |

<sup>a</sup>40 studies, 15,879 patients. Shown are prevalence rates of patients with vascular disease affecting other regional beds identified by angiography.

<sup>b</sup>(?) marks refer to limited data suggesting high prevalence in unexplained advanced chronic kidney disease (CKD) in older subjects and patients with congestive heart failure.

AAA, Abdominal aortic aneurysm; ESKD, end-stage kidney disease.

Modified from de Mast Q, Beutler JJ. The prevalence of atherosclerotic renal artery stenosis in risk groups: a systematic literature review. *J Hypertens.* 2009;27:1333–1240.

### Box 47.2 Clinical features of patients with renovascular hypertension

Syndromes associated with renovascular hypertension

1. Early- or late-onset hypertension (<30 years >50 years)
2. Acceleration of treated essential hypertension
3. Deterioration of renal function in treated essential hypertension
4. Acute renal failure during treatment of hypertension
5. “Flash” pulmonary edema
6. Progressive renal failure
7. Refractory congestive cardiac failure

The above “syndromes” should alert the clinician to the possible contribution of renovascular disease in a given patient. Numbers 5 to 7 are most common in patients with bilateral disease, many of whom are treated as “essential hypertension” until these characteristics appear.

renovascular hypertension are older individuals whose previously well-controlled hypertension deteriorates with an accelerated rise in systolic blood pressure and target injury, such as stroke. Studies from hypertension referral centers in the Netherlands illustrate this. Of 477 patients undergoing

detailed evaluation for RAS because of “treatment resistance,” 107 (22.4%) were identified with RVD (>50% stenosis by angiography).<sup>56</sup> Ambulatory blood pressure recordings indicate exaggerated systolic pressure variability and frequent loss of the circadian pressure rhythm,<sup>62</sup> commonly associated with left-ventricular hypertrophy.<sup>63</sup> Sympathetic nerve traffic recordings indicate heightened adrenergic outflow.<sup>64</sup> A population-based study of 870 subjects older than age 65 indicated that those with RVD had a twofold to threefold increased risk for adverse cardiovascular events during the subsequent 2 years.<sup>65</sup>

**Table 47.2 Clinical features of patients with renovascular hypertension<sup>a</sup>**

| Clinical feature               | Essential HTN (%) | Renovascular HTN (%) |
|--------------------------------|-------------------|----------------------|
| Duration <1 year               | 12                | 24                   |
| Age of onset >50 years         | 9                 | 15                   |
| Family history of HTN          | 71                | 46                   |
| Grade 3 or 4 fundi             | 7                 | 15                   |
| Abdominal bruit                | 9                 | 46                   |
| Blood urea nitrogen > 20 mg/dL | 8                 | 16                   |
| Potassium <3.4 mEq/L           | 8                 | 16                   |
| Urinary casts                  | 9                 | 20                   |
| Proteinuria                    | 32                | 46                   |

<sup>a</sup>Clinical features that differed ( $P < .05$ ) between closely matched groups of 131 patients with essential versus renovascular hypertension taken from the Cooperative Study of Renovascular Hypertension in the 1960s.

HTN, Hypertension.

(From Simon N, Franklin SS, Bleifer KH, Maxwell MH. Clinical characteristics of renovascular hypertension. *JAMA*. 1972;220:1209–1218.) These observations underscore the potential severity of hypertension in candidates for surgery, but none of these features allows clinical discrimination with confidence.

Declining renal function during antihypertensive therapy is a common manifestation of progressive renal arterial disease. Not surprisingly, blood flow and perfusion pressures to the kidney fall distal to a “critical” RAS. This can be worsened by reduction in systemic pressure by any antihypertensive regimen. Reduced GFR during antihypertensive therapy has become particularly common since the introduction of ACE inhibitors and angiotensin receptor blockers. A precipitous rise in serum creatinine soon after starting these agents may occur due to decreases in glomerular hydrostatic pressure and accompanying loss of transcapillary filtration pressure produced by removing the efferent arteriolar vasoconstriction elicited by angiotensin II. This particular “functional” loss of GFR is reversible if detected promptly and should lead the clinician to consider large-vessel RVD when it occurs.<sup>66,67</sup> Clinically important changes in serum creatinine become apparent, mainly when the entire renal mass is affected, such as with bilateral RAS or stenosis to a solitary functioning kidney. Many of these patients tolerate reintroduction of renin-angiotensin blockade when challenged after successful revascularization.<sup>68</sup>

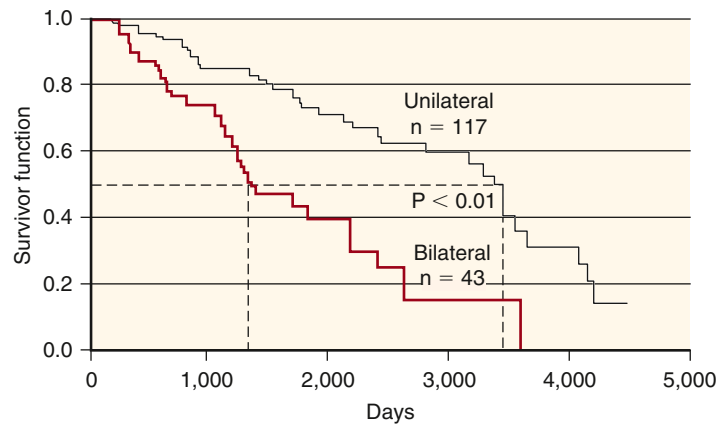
Other syndromes heralding occult RAS are becoming more commonly recognized. Among the most important are rapidly developing episodes of circulatory congestion (“flash” pulmonary edema).<sup>69</sup> This clinical scenario usually arises in patients with hypertension and left-ventricular systolic function, which may be well preserved. Underlying arterial compromise may favor volume retention and resistance to diuretics in such cases. A sudden rise in arterial pressure impairs cardiac function due to rapidly developing diastolic dysfunction. Such episodes tend to be rapid in both onset and resolution. Patients with treatment-resistant congestive cardiac failure, often with reduced arterial

pressures, may also harbor unsuspected RVD. Restoration of renal blood flow in such patients can improve volume control and sensitivity to diuretics with lower risk of azotemia during therapy.<sup>70</sup> A similar sequence of events may produce symptoms of “crescendo” angina from otherwise stable coronary disease.<sup>71</sup> Registry data indicate that patients with episodic pulmonary edema with RVD have substantially increased hospitalization and mortality that can be reduced with successful revascularization.<sup>72,73</sup>

Another clinical presentation of RAS is advanced renal failure, occasionally at end-stage requiring kidney replacement therapy. Studies in patients with bilateral RAS indicate that reduction of systemic pressures to normal levels using sodium nitroprusside can abruptly reduce both renal plasma flow and glomerular filtration rate, indicating that the poststenotic pressures are at critical levels beyond autoregulation (eFig. 47.4).<sup>74</sup> Some estimates suggest that up to 12% to 14% of patients reaching end-stage kidney failure (ESKF) with no other identifiable primary renal disease may have occult, bilateral renal arterial stenosis.<sup>75,76</sup> A more conservative review of U.S. Renal Data Systems data for patients older than age 67 in the United States starting dialysis suggests that identified RVD may be present between 7.1% and 11.1% of patients, although clinicians caring for such patients attributed their renal failure to RVD in only 5.0%.<sup>77</sup> Importantly, patients with rapidly progressive dysfunction and accelerated hypertension have the potential to recover kidney function and a mortality benefit from revascularization.<sup>4,73,78</sup> Such individuals were rarely included in prospective, randomized trials (see later).

The causal role of vascular impairment in producing renal dysfunction is established most firmly when renal revascularization leads to recovery of renal function. Unfortunately, this does not occur commonly.<sup>3,7</sup> A subset of patients with more severe or long-standing vascular occlusion (usually defined as peak systolic velocities >385 cm/sec on duplex ultrasound) manifest overt cortical hypoxia using BOLD MR.<sup>38</sup> Such patients demonstrate more advanced tissue histologic injury with loss of tubular structures on biopsy and interstitial cellular infiltrates consistent with inflammatory injury<sup>38,41</sup> (eFigs. 47.6 and 47.7). These observations support the concept of transition from a primarily “hemodynamic” reduction in kidney function, one that may be improved by restoring blood flow, to inflammatory-mediated injury that no longer predictably recovers after restoration of vessel patency.

The potential benefit of revascularization regarding salvage, or at least stabilization, of renal function is greatest when serum creatinine is <3 mg/dL.<sup>79</sup> Remarkably, RAS can be associated with proteinuria, occasionally to nephrotic levels.<sup>80–82</sup> Proteinuria can diminish or resolve entirely following renal revascularization.<sup>83</sup> Although poststenotic kidneys may lose GFR and tissue volume, the contralateral kidney without stenosis undergoes compensatory hypertrophy. These changes mask the damage to the affected kidney, making overall GFR an unreliable marker of the severity and progression of atherosclerotic RAS.<sup>84,85</sup> Most patients with episodic pulmonary edema have bilateral disease or a solitary kidney. Long-term mortality during follow-up is higher when bilateral disease is present, regardless of whether renal revascularization is undertaken (eFig. 47.7).<sup>86–88</sup>



**eFig. 47.7 Kaplan-Meier survival curve of 160 patients with more than 70% renal artery stenosis managed without revascularization.** Those with bilateral disease had lower survival, primarily due to associated cardiovascular disease. The mean age of death was 79 years. These data underscore the close relationship between the extent of vascular disease and mortality. Less than 10% of these subjects developed advanced kidney disease during follow-up, although long-term survival, even in treated patients, is related to levels of kidney function at the time of intervention. Recent prospective randomized clinical trials indicate that between 16% and 22% of medically treated patients with atherosclerotic renal artery stenosis progress to more advanced renal dysfunction over 3 to 5 years. (From Foster JH, Maxwell MH, Franklin SS, et al. Renovascular occlusive disease: results of operative treatment. *JAMA*. 975;2231:1043–1198; Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al, eds. *Brenner and Rector's: The Kidney*. 10th ed. Philadelphia: Elsevier; 2016:1567–1609.)

## PROGRESSIVE VASCULAR OCCLUSION

Atherosclerosis is a progressive disorder, although individual rates of progression vary widely (Fig. 47.9). Clinical manifestations of RVD depend partly on the severity and extent of vascular occlusion. Zierler reported a 20% rate of RVD progression overall with 7% advancing to total occlusion over 3 years.<sup>89</sup> A later report from the same group using different Doppler velocity criteria suggested higher rates of progressive stenosis.<sup>90</sup> Data from medical treatment trials suggest that progressive occlusion can develop silently in up to 16% of treated subjects.<sup>91</sup> These observations are supported by development of renal endpoints (defined as progressive renal deterioration) ranging between 16% and 22% in recent prospective RCTs (ASTRAL and CORAL).<sup>3</sup>

Most series of medically treated patients indicate that despite evident progression of vascular disease, changes in kidney function are modest and uncommon.<sup>92</sup> Results reported during follow-up of 41 patients managed medically for an average of 36 months before the introduction of ACE inhibitors identified a loss of renal length in 35%, while a significant rise in serum creatinine developed in only 8/41 (19.5%).<sup>92</sup> This conclusion is consistent with long-term studies from Europe in which incidental renal artery lesions were rarely associated with progressive renal failure over more than 9 years of follow-up.<sup>93</sup> These data support the observation that renal artery lesions remain stable in some patients over many years without adverse clinical effects or evident progression,<sup>94</sup> as observed in treatment trials including ASTRAL and CORAL.<sup>95</sup>

## DIAGNOSTIC TESTING FOR RENOVASCULAR HYPERTENSION AND ISCHEMIC NEPHROPATHY

### GOALS OF EVALUATION

It is important that the clinician identify the objectives of evaluating RVD before initiating expensive and sometimes

ambiguous investigations. As with all tests, the reliability and value of diagnostic studies depend heavily on the pretest probability of disease<sup>96</sup> (Box 47.3 and eFig. 47.8). It is essential to consider from the outset exactly what is to be achieved. Is the major goal to exclude high-grade renal artery disease? Is it to exclude bilateral (as opposed to unilateral) disease? Is it to identify stenosis and estimate the potential for clinical benefit from renal revascularization? Is it to evaluate

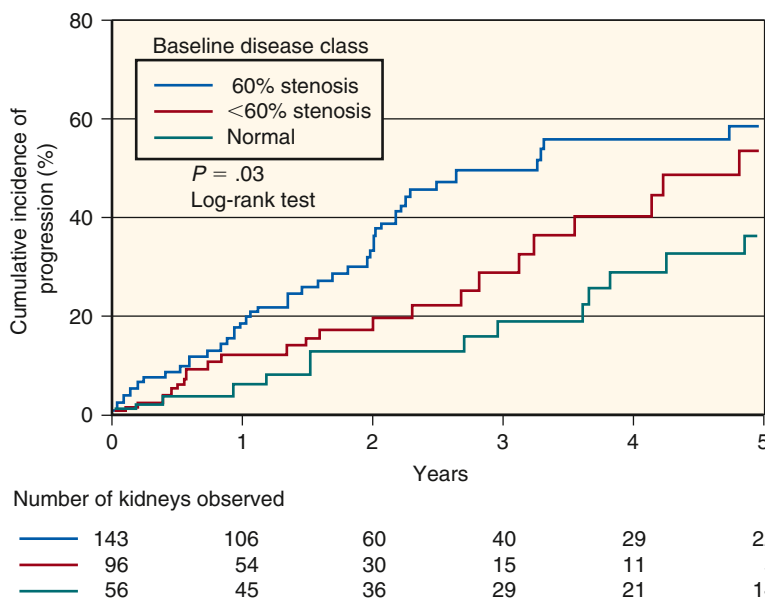
### Box 47.3 Renovascular disease and ischemic nephropathy

#### Goals of diagnostic evaluation

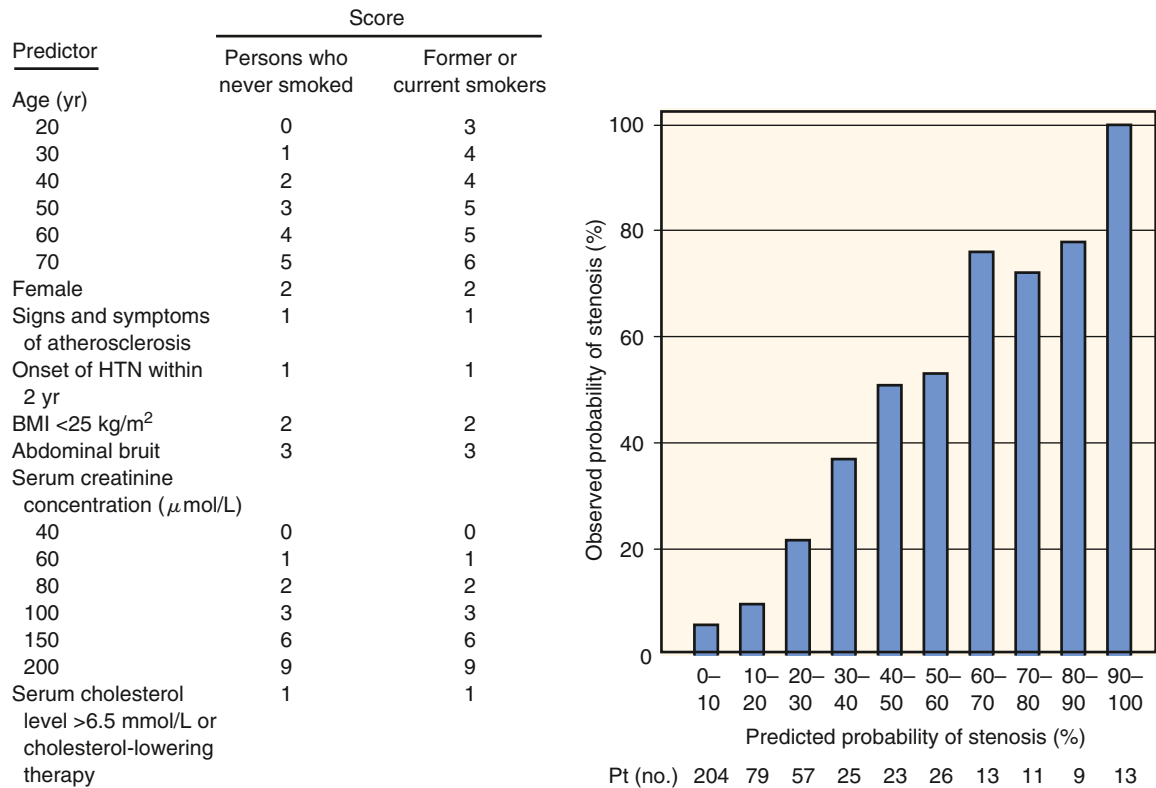
- Establish presence of renal artery stenosis: location and type of lesion
- Establish whether unilateral or bilateral stenosis (or stenosis to a solitary kidney) is present
- Establish presence and function of stenotic and nonstenotic kidneys
- Establish hemodynamic severity of renal arterial disease
- Evaluate progression of vascular occlusion/renal atrophy
- Plan vascular intervention: degree and location of atherosclerotic disease

#### Goals of therapy

- Improved blood pressure control:
  - Prevent morbidity and mortality of high blood pressure
  - Improve blood pressure control and reduce medication requirement
- Preservation of renal function
  - Reduce risk of renal adverse perfusion from use of antihypertensive agents
  - Reduce episodes of circulatory congestion ("flash" pulmonary edema)
  - Reduce risk of progressive vascular occlusion causing loss of renal function: "preservation of renal function"
  - Salvage renal function (i.e., recover glomerular filtration rate)
- Recovery of volume regulation and fluid excretion
  - Reduce circulatory overload and diuretic resistance
  - Reduce episodes of "flash pulmonary edema"



**Fig. 47.9 Cumulative rates of anatomic disease progression in atherosclerotic renal artery stenosis, as measured by renal artery Doppler ultrasound.** During a follow-up period of 5 years, overall progression was 31%, but those with the most severe baseline lesion progressed in 60% of cases. The progression of vascular disease was not closely related to changes in serum creatinine or renal atrophy. (Modified from Radermacher J, Weinkove R, Haller H. Techniques for predicting a favourable response to renal angioplasty in patients with renovascular disease. *Curr Opin Nephrol Hypertens.* 2001;10(6):799–805.)



**eFig. 47.8 Probability of identifying renal artery stenosis based on clinical features.** These data were derived from 477 patients in referral centers for treatment-resistant hypertension (HTN) in the Netherlands. Overall prevalence was 22.4%, illustrating that even in “enriched” patient populations, renovascular disease is not present in the majority. Clinical features allowed selection of patients for testing with a relatively high pretest probability of disease, which affects the validity of testing schemes. (From Krijnen P, van Jaarsveld BC, Steyerberg EW, et al. A clinical prediction rule for renal artery stenosis. *Ann Intern Med.* 1998;129:705–711.)

**Table 47.3 Noninvasive assessment of renal artery stenosis**

| Study                                                                                                                            | Rationale                                                                                            | Strengths                                                                                                                           | Limitations                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Physiologic studies to assess the renin-angiotensin system</b>                                                                |                                                                                                      |                                                                                                                                     |                                                                                                                                                                                 |
| Measurement of peripheral plasma renin activity                                                                                  | Reflects the level of sodium excretion                                                               | Measures the level of activation of the renin-angiotensin system                                                                    | Low predictive accuracy for renovascular hypertension; results influenced by medications and many other conditions                                                              |
| Measurement of captopril-stimulated renin activity                                                                               | Produces a fall in pressure distal to the stenosis                                                   | Enhances the release of renin from the stenotic kidney                                                                              | Low predictive accuracy for renovascular hypertension; results influenced by many other conditions                                                                              |
| Measurement of renal vein renin activity                                                                                         | Compares renin release from the two kidneys                                                          | Lateralization predictive of improvement in blood pressure with revascularization                                                   | Nonlateralization has limited predictive power of the failure of blood pressure to improve after revascularization; results influenced by medications and many other conditions |
| <b>Perfusion studies to assess differential renal blood flow</b>                                                                 |                                                                                                      |                                                                                                                                     |                                                                                                                                                                                 |
| Captopril renography with technetium <sup>99m</sup> Tc mertiatide ( <sup>99m</sup> Tc MAG3)                                      | Captopril-mediated fall in filtration pressure amplifies differences in renal perfusion              | Normal study excludes renovascular hypertension                                                                                     | Multiple limitations in patients with advanced atherosclerosis or creatinine >2.0 mg/dL (177 μmol/L)                                                                            |
| Nuclear imaging with technetium mertiatide or technetium-labeled pentetic acid (DTPA) to estimate fractional flow to each kidney | Estimates fractional flow to each kidney                                                             | Allows calculation of single-kidney glomerular filtration rate                                                                      | Results may be influenced by other conditions (e.g., obstructive uropathy)                                                                                                      |
| <b>Vascular studies to evaluate the renal arteries</b>                                                                           |                                                                                                      |                                                                                                                                     |                                                                                                                                                                                 |
| Duplex ultrasonography                                                                                                           | Shows the renal arteries and measures flow velocity as a means of assessing the severity of stenosis | Inexpensive; widely available: suitable for sequential measurement to follow disease progression and/or restenosis                  | Heavily dependent on operator's experience; less useful than invasive angiography for the diagnosis of fibromuscular dysplasia and abnormalities in accessory renal arteries    |
| Magnetic resonance angiography                                                                                                   | Shows the renal arteries and perirenal aorta                                                         | Not nephrotoxic, but concerns for gadolinium toxicity exclude use in GFR <30 mL/min/1.73 m <sup>2</sup> ; provides excellent images | Expensive; gadolinium excluded in renal failure, unable to visualize stented vessels                                                                                            |
| Computed tomographic angiography                                                                                                 | Shows the renal arteries and perirenal aorta                                                         | Provides excellent images; stents do not cause artifacts                                                                            | Expensive, moderate volume of contrast required, potentially nephrotoxic                                                                                                        |

Modified from Safian RD, Textor SC. Medical progress: renal artery stenosis. *N Engl J Med*. 2001;344:431–442 with permission.

the role of RVD in explaining deteriorating renal function? The specific approach to diagnosis will differ depending on which of these is the predominant clinical objective.

## IMAGING OF THE RENAL VASCULATURE

Identification of ARVD fundamentally requires some form of imaging the renal vasculature. Advances in Doppler ultrasound, radionuclide imaging, magnetic resonance arteriography (MRA), and computed tomography (CT) angiography continue to improve renovascular imaging. The details of these methods are discussed further in [Chapter 24](#). What follows is

a discussion of some of the specific merits and limitations of each modality as they apply to application in renovascular hypertension and ischemic nephropathy<sup>96</sup> ([Table 47.3](#)).

Current practice favors limiting invasive arteriography to the occasion of endovascular intervention (e.g., stenting and/or angioplasty). While angiography remains the “gold standard” for evaluation of the renal vasculature, its invasive nature, potential hazards, and cost make it most suitable for those in whom intervention is planned. As a result, most clinicians favor preliminary noninvasive studies. Arterial angiography may be warranted to establish the presence of transstenotic pressure gradients, as recommended in treatment trials.<sup>97,98</sup>

## NONINVASIVE IMAGING

### DOPPLER ULTRASOUND OF THE RENAL ARTERIES

Duplex interrogation of the renal arteries provides measurements of localized velocities of blood flow and characteristics of renal tissue. In many institutions, this provides an inexpensive means for measuring vascular occlusive disease at sequential time points to establish the diagnosis of RVD and monitor its progression. After renal revascularization, Doppler studies are commonly used to monitor restenosis and target vessel patency<sup>99,100</sup> (eFig. 47.9A and B) (see Chapter 24). Its main drawbacks relate to the difficulties of obtaining adequate studies in obese patients, operator dependency, and time required. These factors vary considerably between institutions. It is important to remember that blood velocity is not the same as blood flow. The volume of unit passing per unit of time (aka blood flow) can be impuned with associated M-mode measurements of internal vessel diameter, but it is a distinct concept from that of blood velocity. Importantly, when reasonable blood flow is present, the primary criteria for renal artery studies are a peak systolic velocity above 180 cm/sec and/or a relative velocity above 3.5 as compared with the adjacent aortic flow.<sup>101,102</sup> Using these criteria, sensitivity and specificity with angiographic estimates of lesions exceeding 60% can surpass 90 and 96%, respectively.<sup>101</sup> Increasing the threshold for peak systolic velocities reduces the rate of false-positive estimates of stenosis. Because the correlation between velocity and degree of stenosis is only approximate, clinical trials such as CORAL raised the peak systolic velocity threshold to 300 cm/s.<sup>101</sup> This seems warranted, particularly when the risk of overdiagnosis of renal arterial lesions is high, as in the STAR trial, in which 18/64 patients assigned to stenting were found not to have significant RVD at the time of angiography despite noninvasive estimates to the contrary.<sup>103</sup>

Segmental waveforms within the arcuate vessels in the renal hilum can provide additional information. Damping of these waveforms, labeled as “parvus” and “tardus,” are indirect signs of upstream vascular occlusive phenomena.<sup>104</sup> Some authors challenge the use of angiographic estimates of stenosis as representing a “gold standard” altogether.<sup>105</sup> These authors argue that Doppler velocities correlate highly ( $R=0.97$ ) with a truer estimate of vascular occlusion, specifically luminal stenosis determined by intravascular or endovascular ultrasound.

Additional studies emphasize the potential for Doppler ultrasound to characterize small vessel flow characteristics within the kidney.<sup>106</sup> The “resistive index” provides an estimate of the relative flow velocities in diastole and systole. Normally, the velocity of blood in the measured vessels when defining the “resistive index” is not highly pulsatile. When diastolic blood velocity falls 50% relative to systolic blood velocity, the “resistive index” is 80. In this setting the blood velocity is highly pulsatile during the cardiac cycle. When diastolic blood velocity falls further than 50% relative to systolic blood velocity, then the “resistive index” increases above 80. In a study of 138 patients with RAS, a resistive index above 80 provided a predictive tool for identification of parenchymal renal disease that did not respond to renal revascularization (eFig. 47.10). A resistive index <80 was associated with >90% favorable blood pressure response and stable or improved renal function

following renal revascularization. The authors emphasize that accurate predictive power depended on using the highest resistive index observed, even when present in the nonstenotic kidney. A subsequent study of 215 subjects with mean preintervention serum creatinine levels of 1.51 mg/dL failed to confirm the predictive value of resistive index measurements.<sup>107</sup> In this series, preintervention level of serum creatinine itself was the strongest predictor of improved renal function. Most clinicians agree that detecting a low resistive index indicates a well-preserved vasculature within the kidney with improved likelihood of recovering or stabilizing after vascular intervention.<sup>108,109</sup>

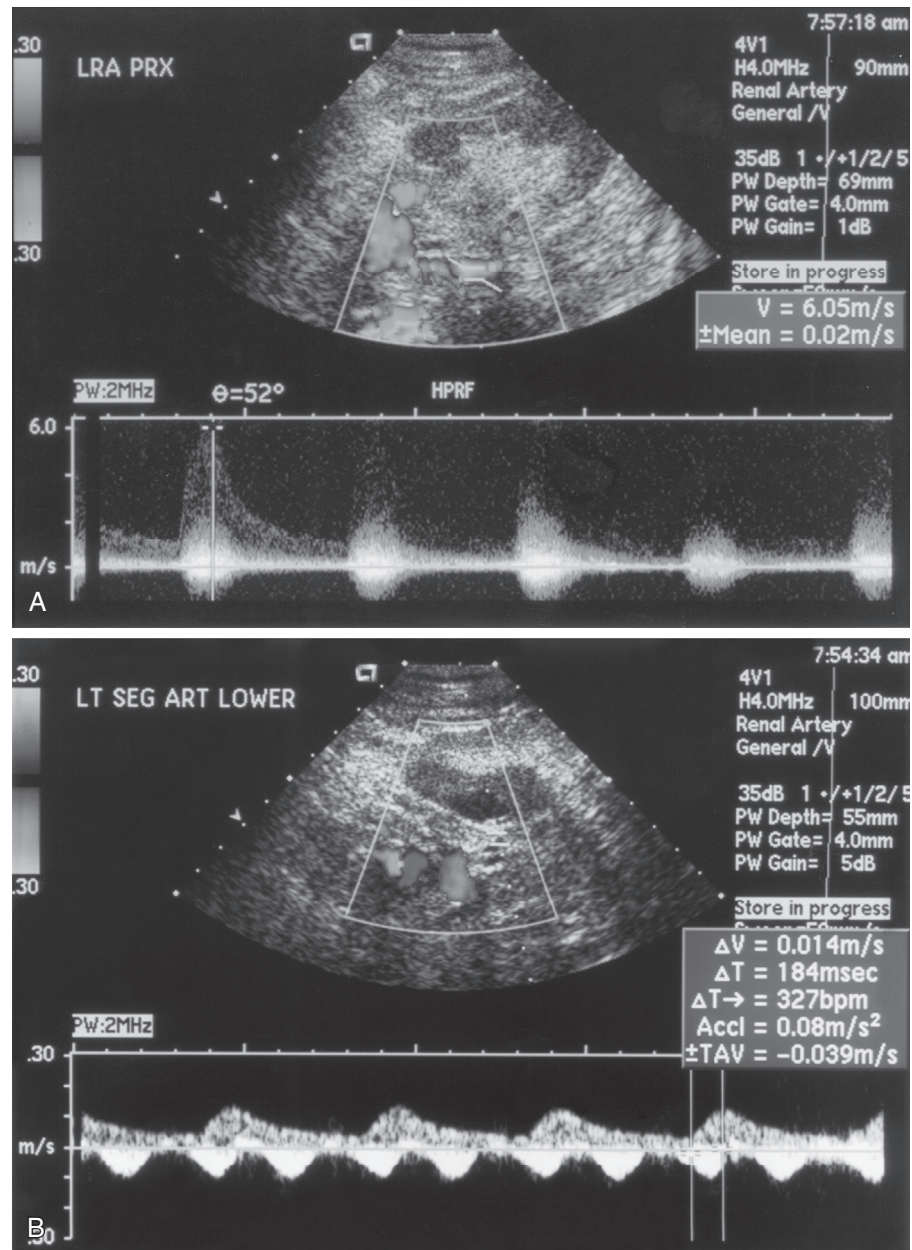
### COMPUTED TOMOGRAPHY ANGIOGRAPHY

CT angiography using “spiral” or “helical” and/or multiple detector scanners and intravenous contrast can provide excellent images of both kidneys and the vascular tree. Resolution and reconstruction techniques render this modality capable of identifying smaller vessels, vascular lesions, and parenchymal characteristics including malignancy and stones<sup>110</sup> (eFig. 47.10). When used for detection of RAS, CT angiography agrees well with conventional arteriography (correlation 95%) and sensitivity may reach 98% and specificity 94%.<sup>111</sup> CT also provides excellent accuracy regarding evaluation of in-stent restenosis,<sup>112</sup> and evolving quantitative three-dimensional image analysis may improve on intraarterial methods.<sup>111</sup> Remarkably, patients with atherosclerotic RAS tolerate contrast from CT imaging without evident toxicity, despite advanced age and low GFR.<sup>113</sup> Limitations include reduced visibility of vessel lumens in the presence of substantial calcium deposition and/or previous stent placement. The results of such studies reinforce the importance of careful patient selection for study and establishing in advance exactly the purpose for which imaging is being undertaken.

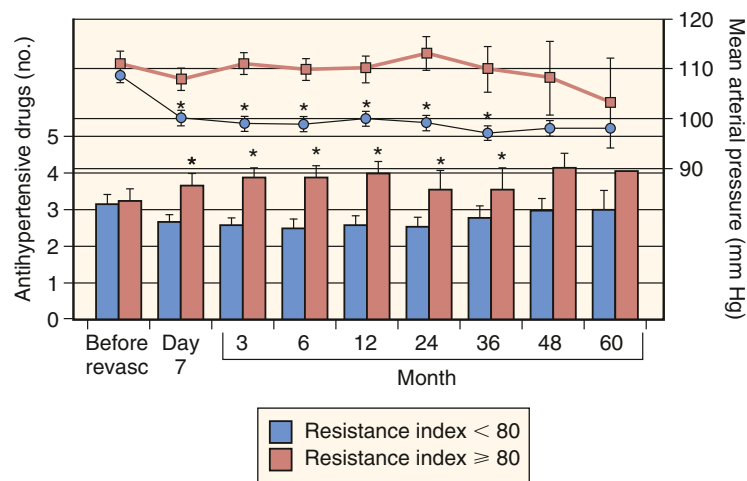
### MAGNETIC RESONANCE ANGIOGRAPHY

Gadolinium-enhanced images of the abdominal and renal vasculature have been a mainstay of evaluating RVD in many institutions.<sup>110,114</sup> Comparative studies indicate that sensitivity ranges from 83% to 100% and specificity from 92% to 97% in RAS.<sup>115,116</sup> Meta-analyses of published literature including 998 subjects support more than 97% sensitivity using gadolinium-enhanced imaging.<sup>117</sup> The nephrogram obtained from gadolinium filtration provides an estimate of relative function and filtration, as well as parenchymal volume. Quantitative measurement of parenchymal volume determined by MR imaging appears to correlate closely with isotopically determined single-kidney GFR in some institutions<sup>118</sup>. Since 2006, concerns about the potential for gadolinium-based contrast to produce nephrogenic systemic fibrosis have reduced utilization of contrast-enhanced MR for patients with impaired kidney function in the United States.<sup>119</sup> This topic is further discussed in Chapter 24 as newer generation forms of gadolinium are gaining wider acceptance at reduced levels of renal function.

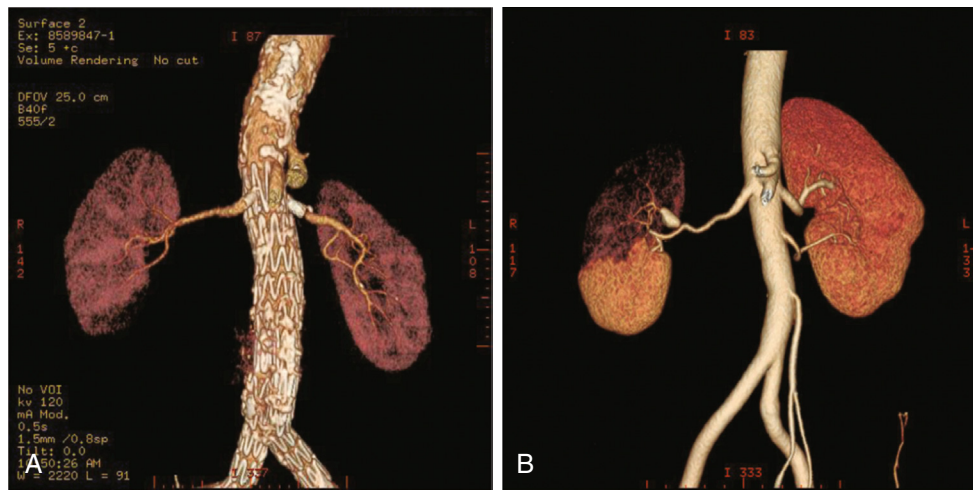
Technologic advances allow high-resolution vascular MR imaging without contrast in many patients. An example of MR angiography without contrast is shown in Fig. 47.11. Drawbacks include expense and the tendency to overestimate the severity of lesions, which in fact appear as a signal void.<sup>110,120</sup> The limits of resolution with current magnet strength of MR imaging instrumentation make



**eFig. 47.9** (A) Duplex ultrasound velocity measurement in a patient with high-grade renal artery stenosis affecting the proximal left renal artery (*LRA PRX*). Peak systolic velocities reach 605 cm/s (6.05 m/s), well above the normal upper limit of 180 cm/s. (B) Segmental arterial branch ultrasound in the distal segmental renal arteries demonstrates “parvus” and “tardus” dampening of the signal characteristic of poststenotic waveforms. The utility of these measurements depends on the ability to obtain reliable identification of vessel segments and the skills of the operator. Once the location of a vascular lesion is known, subsequent studies can be performed more easily to track progression of vascular occlusion, restenosis, and/or the results of endovascular intervention.



**eFig. 47.10 Outcome of revascularization as measured by mean arterial blood pressure and number of antihypertensive agents in 138 patients with renal artery stenosis.** These patients were divided into groups with ultrasound-determined resistive index above 80 and those lower than 80 in the most severely affected kidney. The authors indicate that a high resistive index reflects intrinsic parenchymal and small vessel disease in the kidney that does not improve after revascularization. Those with lower indices had both lower blood pressures during follow-up and lower antihypertensive medication requirements. (From Radermacher J, Chavan A, Bleck J, et al. Use of Doppler ultrasonography to predict the outcome of therapy for renal-artery stenosis. *N Engl J Med.* 2001;344:410–417.)



**Fig. 47.10 Computed tomography (CT) angiograms illustrating reconstructed views of complex vascular disease.** (A) Aortic endovascular stent extending beyond the origins of the renal arteries. Renal artery stents have been placed through the aortic graft to restore blood flow, although the nephrograms demonstrate patchy defects consistent with small vessel occlusion and/or atheroembolic events. (B) Illustrates a CT angiogram with a small aneurysm of the right renal artery that has produced segmental infarction, leading to accelerated hypertension. Although CT angiography requires contrast, current multidetector CT studies allow excellent image resolution at rapid acquisition and less contrast exposure than ever before.



**Fig. 47.11 Magnetic resonance (MR) angiogram reconstructed without gadolinium contrast.** After warnings about the association of gadolinium with nephrogenic systemic fibrosis, contrast MR is less commonly used for atherosclerotic renovascular imaging than before. Newer techniques can allow excellent vascular imaging without contrast as shown here.

detection of small accessory vessels and quantitating fibromuscular lesions difficult. Both are improving with newer generations of scanners. High-spatial resolution three-dimensional contrast-enhanced MR scanners provide up to 97% sensitivity and 92% specificity for renal artery stenotic lesions.<sup>121</sup> Signal degradation in the presence of metallic stents renders MRA unsuitable for follow-up studies after endovascular procedures in which stents are used.

### CAPTOPRIL RENOGRAPHY

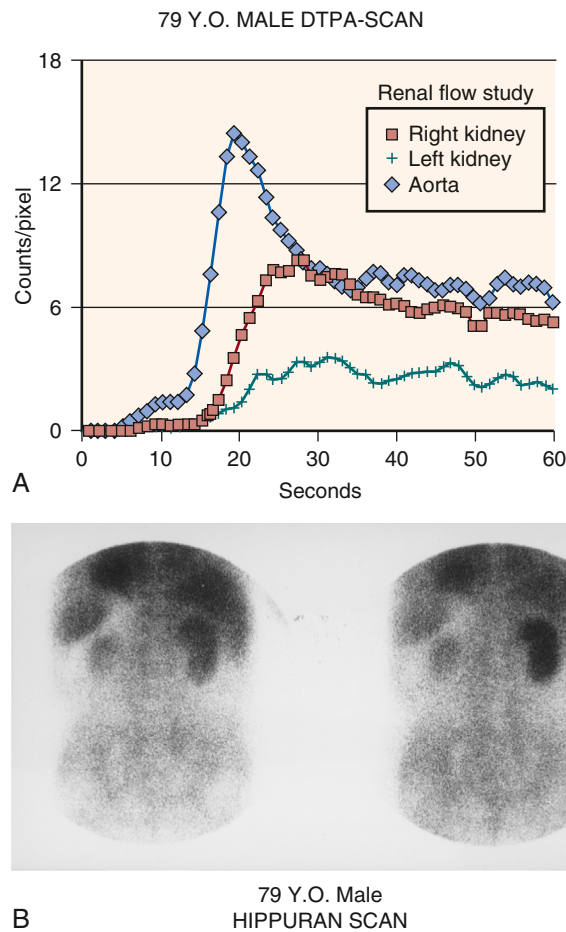
Imaging the kidneys before and after administration of an ACE inhibitor (e.g., captopril) can provide a functional assessment of the change in blood flow and GFR to the kidney related both to changes in arterial pressure and removal of the efferent arteriolar effects of angiotensin II. The most used radiopharmaceuticals are technetium-99 (99mTc) diethylene triamine pentaacetic acid (DTPA) and 99mTc-mercaptoacetyl triglycine (MAG3). The latter agent has clearance characteristics similar to hippuran and is often taken as reflecting renal plasma flow.<sup>122,123</sup>

Both provide information regarding size and filtration of both kidneys, and the change in these characteristics after inhibition of ACE allows inferences regarding the dependence of glomerular filtration on angiotensin II. Renography, however, only provides functional information and no direct anatomic information (i.e., the location of renal artery disease, the number of renal arteries, or associated aortic and/or ostial disease) (eFig. 47.11A and B). Prospective studies of RVD from the Netherlands did observe changes in the renogram during follow-up but did not find captopril renography predictive of angiographic findings or outcomes of revascularization.<sup>91</sup>

Changes in split renal function indicate that stenotic kidneys regain GFR after revascularization, sometimes with a decrement in contralateral GFR, thereby maintaining overall kidney function unchanged.<sup>124</sup> Currently, radionuclide studies have a role in identifying the relative contributions of each kidney to overall GFR—often as part of evaluation for residual renal function after nephrectomy.<sup>125</sup>

### INVASIVE IMAGING

Intraarterial angiography currently remains the “gold standard” for definition of vascular anatomy and stenotic lesions in the kidney, often performed at the time of a planned intervention, such as endovascular angioplasty and/or stenting. Endovascular procedures for such lesions should be confined to individuals with strong indications for renal revascularization, as even ardent advocates of catheter-based intervention have suggested. Contrast toxicity remains an issue with conventional iodinated agents given intraarterially.<sup>126</sup> With developments in digital subtraction angiography, CO<sub>2</sub> gas has been used as an alternative contrast agent. Because of its lack of renal toxicity and allergic potential, CO<sub>2</sub> is preferred as contrast agent by some centers, especially in patients with renal failure or contrast allergy.



**eFig. 47.11 (A and B) Isotope renography in a patient with unilateral renal artery stenosis.** (A) A  $^{99m}\text{Tc}$ -DTPA scan demonstrates delayed circulation and excretion of isotope on the *left*. (B) A hippuran scan (now replaced with  $^{99m}\text{Tc}$ -mercaptoacetyltriglycine) provides a renogram demonstrating a small kidney with impaired renal function on the affected side. Radionuclide scans provide a comparative estimate of function from each kidney that may facilitate selection of intervention including the potential effect of nephrectomy. *DTPA*, Diethylenetriamine pentaacetic acid.

Intravascular ultrasound procedures have been undertaken using papaverine to evaluate “flow reserve” beyond stenotic lesions.<sup>127</sup> Various reports indicate that either a reduction or preserved flow reserve may identify a poststenotic kidney that may benefit from successful revascularization.<sup>127,128</sup> Measurements using currently available low-profile wire probes indicate a relationship between pressure gradients and activation of the renin-angiotensin system.<sup>14</sup> Outcomes of patients with translesional pressure gradients measured after vasodilation suggest that measurement of “hyperemic systolic gradient” above 21 mm Hg (defined after vasodilation with papaverine) most accurately predicts high-grade stenosis (average 78% by intravascular ultrasound) and a beneficial response of blood pressure after stenting.<sup>129</sup> The latter observation and increasing reliability of technical measurements underscore the value of measuring gradients to establish a hemodynamic role for vascular lesions of marginal severity.

### FUNCTIONAL STUDIES OF THE RENIN-ANGIOTENSIN SYSTEM

Efforts for many years used activation of the renin-angiotensin system as a marker of underlying renovascular hypertension.<sup>108,130</sup> While these studies are promising when performed in patients with known renovascular hypertension, they have lower performance when applied to wider populations. Plasma renin activity is sensitive to changes in sodium intake, volume status, and renal function and is especially sensitive to varied antihypertensive medications.

Measurement of renal vein renin levels obtained by sampling renal vein and inferior vena cava blood individually had been widely applied in planning surgical revascularization for hypertension. The level obtained from the vena cava is comparable with the arterial levels into each kidney and allows estimation of the contribution of each kidney to total circulating levels of plasma renin activity. Lateralization is defined usually as a ratio exceeding 1.5 between the renin activity of the stenotic kidney and nonstenotic kidney.<sup>131</sup> In general, the greater the degree of lateralization, the more probable that blood pressure will benefit from surgical or other revascularization.<sup>132</sup> These observations have been supported by studies of renal vein measurements before consideration of nephrectomy for refractory hypertension and advanced RVD.<sup>125</sup> A review of more than 50 studies of renal vein renin measurements indicated that when lateralization could be demonstrated, clinical benefit regarding blood pressure control could be expected in more than 90% of cases.<sup>133</sup> Failure to demonstrate lateralization, however, still was associated with significant benefit in more than 50% of cases.<sup>86</sup> In cases for which it is important to establish the degree of pressor effect of a specific kidney or site before nephrectomy, measurement of renal vein renin levels can provide strong supportive evidence.<sup>125</sup>

### STUDIES OF INDIVIDUAL RENAL FUNCTION

Serum creatinine and estimates of GFR measure total combined renal excretory function and do not address changes within each kidney. An early body of literature addressed the potential for individual “split” renal function studies to establish the functional importance of each

kidney in RVD.<sup>134</sup> These split function studies classically utilized separate ureteral catheters to allow individual urine collection for measurement of single-kidney GFR, renal blood flow, sodium excretion, concentrating ability, and the response to blockade of angiotensin II.

Separate renal functional measurements can be obtained less invasively with radionuclide techniques, as noted earlier. Serial measurements allow more precise identification of progressive “ischemic” injury to the affected kidney in unilateral renal artery disease than can be determined from overall glomerular filtration rate.<sup>135</sup> The poststenotic kidney is susceptible to gradual loss of filtration that is accompanied by a rise in blood flow and filtration in the contralateral kidney, both of which can partially reverse after restoring blood flow.<sup>84</sup>

## MANAGEMENT OF RENAL ARTERY STENOSIS AND ISCHEMIC NEPHROPATHY

The past decade has been associated with a major shift from primary renal revascularization to reliance on medical antihypertensive therapy for atherosclerotic RVD. This “pendulum swing” may be excessive in some cases<sup>1,136</sup>; however, eventually exposing patients to high-risk clinical sequelae and advanced renal failure might be prevented. Considering the array of potential interventions and the complexity of these patients, clinicians need to formulate a clear set of therapeutic goals for each individual. In most cases, management of the patient with RVD requires integrated pharmacologic management of blood pressure and cardiovascular risk over the long term, combined with optimal timing of renal revascularization for those with high-risk manifestations and/or disease progression. The objective of this section is to provide a framework by which to plan a balanced approach to patients with unilateral or bilateral RVD.

### MEDICAL THERAPY OF RENOVASCULAR DISEASE

The goals of therapy are summarized in [Box 47.3](#). Foremost among these is the National High Blood Pressure Education Program statement: “The goal of treating patients with hypertension is to prevent morbidity and mortality associated with high blood pressure ...”.<sup>137</sup> This task may include efforts to simplify or potentially eliminate long-term antihypertensive drug therapy, particularly in younger individuals with fibromuscular disease. A further goal is to preserve kidney function and prevent loss of kidney function related to impaired renal blood flow. In some instances, renal revascularization is undertaken to allow improved management of salt and water balance in patients with congestive cardiac failure. This may allow safer use of diuretic agents and ACE inhibitor/ARB classes of medication in patients with critical renal artery lesions affecting the entire renal mass. Several prospective, randomized controlled trials have been undertaken since 2005. While each has limitations, results of medical therapy over 2 to 5 years have been equivalent to those observed with medical therapy plus renal revascularization with relatively few treatment crossovers ([Table 47.4](#); see also [eTables 47.1](#) and [47.2](#)).<sup>95</sup> If blood pressure can be well controlled and kidney function remains stable, moving forward with costly and potentially hazardous imaging and/or vascular intervention procedures

**Table 47.4** Prospective, randomized clinical trials for renal function and/or cardiovascular outcomes comparing medical therapy versus renal artery stenting

| <b>Trials</b>                                  | <b>Number</b>           | <b>Population</b>                                                                                                                              | <b>Inclusion criteria</b>                                                                                                                                                                          | <b>Exclusion criteria</b>                                                                                                                     | <b>Outcomes</b>                                                                                                                                                                                                  |
|------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STAR<br>(2009)<br>10 centers<br>f/up 2 years   | Med Tx:76<br>PTRA: 64   | Patients with impaired renal function, ostial ARVD detected by various imaging studies and stable blood pressure on statin and aspirin         | ARVD >50%<br>Creatinine clearance <80 mL/min/1.73 m <sup>2</sup><br>Controlled blood pressure 1 month before inclusion                                                                             | Kidney <8 cm and renal artery diameter <4 mm, eCrCl <15 mL/min per 1.73 m <sup>2</sup> , DM with proteinuria (>3 g/d), malignant hypertension | No difference in GFR decline (primary endpoint $\geq 20\%$ change in clearance), but many did not undergo PTRA due ARVD <50% on angiography<br>Serious complications in the PTRA group<br>Study was underpowered |
| ASTRAL<br>(2009)<br>57 centers<br>f/up 5 years | Med Tx:403<br>PTRA: 403 | Patients with uncontrolled or refractory hypertension or unexplained renal dysfunction with unilateral or bilateral ARVD on statin and aspirin | ARVD substantial disease suitable for endovascular and patient's doctor uncertainty of clinical benefit from revascularization                                                                     | High likelihood of PTRA in <6 mo<br>Without ARVD, previous ARVD<br>PTRA<br>FMD                                                                | No difference in BP, renal function, mortality, CV events<br>(primary endpoint: 20% reduction in the mean slope of the reciprocal of the serum creatinine level)<br>Substantial risk in the PTRA group           |
| CORAL<br>(2013)<br>109 centers<br>f/up 5 years | Med Tx:480<br>PTRA:467  | Hypertension 2 or more antihypertensives or CKD stage $\geq 3$ with ARVD with unilateral or bilateral disease on statin                        | SBP >155 mm Hg, at least 2 drugs<br>ARVD >60%<br>Subsequent changes included that the SBP >155 mm Hg for defining systolic hypertension was no longer specified as long as patient had CKD stage 3 | FMD Creatinine >4.0 mg/dL<br>kidney length <7 cm and use of >1 stent                                                                          | No difference in death from CV or renal causes. Modest improvement of SBP in the stented group<br>Total 26 complications (5.5%)                                                                                  |

ARVD, Atherosclerotic renovascular disease; CKD, chronic kidney disease; CV, cardiovascular; eCrCl, estimated creatinin clearance; FMD, fibromuscular dysplasia; PTRA, percutaneous renal artery angioplasty; RVD, renovascular disease; SBP, systolic blood pressure; Tx, therapy. (Modified from Bhalla V, Textor SC, Beckman JA, et al. Revascularization for renovascular disease: a scientific statement from the American Heart Association. *Hypertension*. 2022;79(8):e128–e143.

may not be justified. These measures must be considered within the entire context of patient management over time.

#### MANAGEMENT OF UNILATERAL VERSUS BILATERAL RENAL ARTERY STENOSIS

“Bilateral” in this context refers to circumstances when the entire functional renal mass is affected by vascular occlusion either by bilateral stenoses or stenosis to a solitary functioning kidney. Not only are the putative mechanisms related to blood pressure and volume control different in the presence of a nonstenosed, functioning contralateral kidney with unilateral disease (as described earlier), but the potential hazards of intervention and/or medical therapy differ. Patient survival is reduced in patients with

bilateral disease or stenosis to a solitary functioning kidney. Progressive arterial disease in this group also poses the most immediate hazard of declining renal function.

#### MANAGEMENT OF UNILATERAL RENAL ARTERY STENOSIS

Most patients with atherosclerotic renal artery disease have preexisting hypertension.<sup>138</sup> As a result, many are exposed to antihypertensive drug therapy before identification of the lesion and may be well controlled with only moderate medication use.<sup>139</sup> Such patients commonly come to clinical attention when recognizable progression occurs.

**eTable 47.1 Summary of observational series in renal revascularization****T1 outcomes of renal artery angioplasty, ±stent placement; hypertension**

|                                                | <b>Cured<sup>a</sup></b>          | <b>Improved</b>                   | <b>No Change</b>                 |
|------------------------------------------------|-----------------------------------|-----------------------------------|----------------------------------|
| 14 series, 678 patients, 98% technical success | Weighted mean, 17%; range, 3%–68% | Weighted mean, 47%; range, 5%–61% | Weighted mean, 36% range, 0%–61% |
| Renovascular hypertension, 472 patients,       | 12%                               | 73%                               | 15%                              |

**T1 renal artery stents: effect on renal function<sup>b</sup> in azotemic patients**

|                                                              | <b>“Improved”</b>                  | <b>“Stabilized”</b>                | <b>Worse</b>                       |
|--------------------------------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 14 series; reporting “impaired renal function”; 496 patients | Weighted mean, 30%; range, 10%–41% | Weighted mean, 42%; range, 32%–71% | Weighted mean, 29%; range, 19%–34% |
| Ischemic nephropathy; 469 patients                           | 41%                                | 37%; no change                     | 22%                                |

<sup>a</sup>Cured, blood pressure at normal levels, no medication; improved, blood pressure lower by at least 10mm Hg, fewer medications; no change, same blood pressure and/or same medications.

<sup>b</sup>Renal function—variably defined as improved, persistent fall in creatinine, variable from 0.3 to 1.0 mg/dL or more; worse, variable, but persistent increase in creatinine 0.3 to 1.0 mg/dL or more.

Modified from Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al, eds. *Brenner and Rector's: The Kidney*. 10th ed. Philadelphia: Elsevier; 2016:1567–1609.

**eTable 47.2** Prospective randomized trials of medical therapy versus angioplasty for hypertension<sup>a</sup>

| Source                                                                                                               | Inclusion, blood pressure measurement                                                                                                                                        | Blood pressure outcome (mm Hg)                                                                                                          | Renal outcome (mm Hg)                                                                                                                                                                   | Comments                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SNRAS, Webster et al., 1998 <sup>127</sup> ; <i>n</i> = 55 (unilateral = 27); <i>n</i> = 135 eligible; RAS > 50%     | DBP ≥95 mm Hg, 2 drugs; exclusion: CVA, MI within 3 mo; creatinine >500 μmol/L; BP, random-zero device; no ACE inhibitors allowed                                            | Unilateral—PTRA, 173/95 Med Rx, 161/88<br>Bilateral—PTRA, 152/83; Med Rx, 171/91;<br><i>P</i> < .01                                     | Creatinine (μmol/L), bilateral PTRA, 188 Med Rx, 157<br>Unilateral PTRA, 144 Med Rx, 168                                                                                                | “... unable to demonstrate any benefit in respect of renal function or event-free survival” (F/U 40 mo)                                                                                                                                                                          |
| EMMA: Plouin et al., 1998 <sup>240</sup> ; <i>n</i> = 49 (unilateral RAS only); RAS >75% or >60%, lateralizing study | Age <75 years; normal contralateral kidney; exclusion: malignant HTN CVA, CHF, MI within 6 mo; BP, automated sphygmomanometer, ABPM at 6 mo                                  | PTRA, 140/81<br>Med Rx, 141/84<br>No. drugs (DDD)—PTRA, 1.0<br>Med Rx, 1.78, <i>P</i> < .01<br>Crossover to PTRA: 7/26 (27%)            | Creatinine Clearance mL/min: (6 mo)<br>PTRA: 77<br>Med Rx: 74<br>Renal artery occlusion: PTRA: 0<br>Med Rx: 0                                                                           | “BP levels and the proportion of patients given antihypertensive treatment were similar 1 year after randomization in the control and angioplasty groups, confirming that the BP-lowering effect of angioplasty in the short and medium terms is limited in atherosclerotic RAS” |
| DRASTIC: Van Jaarsveld, et al., 2000 <sup>133</sup> ; <i>n</i> = 106 ASO RAS >50%                                    | Resistant: 2 drugs; DBP >95 mm Hg or creatinine rise with ACEI; exclusion: creatinine ≥2.3 mg/dL; solitary kidney, total occlusion—kidney <8 cm; BP, automated oscillometric | BP outcomes at 3 mo—PTRA, 169/89<br>Med Rx, 163/88 at 12 mo—PTRA, 152/84<br>Med Rx: 162/88<br>No. of drugs, 1.9 vs. 2.4, <i>P</i> < .01 | Creatinine Cl (3 mo) mL/min:<br>PTRA: 70<br>Med Rx: 59 ( <i>P</i> = .03)<br>Abnormal renograms PTRA: 36%<br>Med Rx: 70% ( <i>P</i> = .002)<br>Renal artery occlusion PTRA 0<br>Med Rx 8 | “In the treatment of patients with hypertension and renal artery stenosis, angioplasty has little advantage over antihypertensive drug therapy.”                                                                                                                                 |

<sup>a</sup>In patients with atherosclerotic renal artery stenosis.

ABPM, Overnight ambulatory reading; ACEI, angiotensin-converting enzyme inhibitor; ASO, atherosclerotic occlusive disease; BP, blood pressure; CHF, congestive heart failure; CVA, cerebrovascular accident; DBP, diastolic blood pressure; DDD, defined daily doses; DRASTIC, Dutch Renal Artery Stenosis Intervention Cooperative; EMMA, Essai Multicentrique Medicaments vs. Angioplastie; F/U, follow-up; HTN, hypertension; Med Rx, medical therapy; MI, myocardial infarction; *n*, number of patients; PTRA, percutaneous renal artery angioplasty; RAS, renal artery stenosis; SNRAS, Scottish and Newcastle Renal Artery Stenosis Collaborative Group; Tx, therapy.

Whether long-term treatment of unilateral RVD with antihypertensive drug therapy poses a long-term hazard to poststenotic kidney function remains unclear. An important feature of renin-angiotensin system blockade is the specific reduction of angiotensin-dependent efferent arteriolar resistance sufficient to lower transcapillary filtration pressures. This can occur despite preserved or augmented blood flow to the glomerulus (eFig. 47.12A and B).<sup>140,141</sup> The fall in GFR due to lower transcapillary filtration pressures heralds approaching critical vascular compromise before blood flow itself is reduced.<sup>140</sup>

The contralateral kidney in unilateral RVD usually supports overall GFR despite reduced filtration in the stenotic kidney. Therefore changes in overall GFR may be undetectable due to increased filtration in the contralateral kidney.<sup>84</sup> Depending on the circumstances, loss of one kidney might be acceptable if one can ensure the patient that the remaining kidney has adequate function and blood supply. The fall in GFR from loss of a poststenotic kidney may represent a loss of GFR analogous to that following nephrectomy for malignancy or donating a kidney for renal transplantation. In such instances, the long-term hazard to the remaining kidney is small, although not negligible.<sup>142-144</sup> Hence the loss of one kidney may pose an acceptable clinical hazard if overall GFR is adequate. Subsequent follow-up to monitor and preserve the condition of the remaining kidney is essential.

## MANAGEMENT OF BILATERAL RENAL ARTERY STENOSIS

Some degree of stenosis is often present in the contralateral kidney. When both kidneys have reduced blood flow, overall renal function is at greater risk. Procedures for treating hypertension and CKD in this setting should account for greater hazards to both kidney function and circulatory control. Thresholds for restoring blood flow and maintaining clinical vigilance should therefore be tightened.

Most patients with bilateral stenosis can be treated medically. Thousands of patients with marginal arterial pressures and clinical heart failure have been treated with a variety of ACE inhibitors and ARBs.<sup>140</sup> These patients are at high risk for undetected renal artery lesions as part of the atherosclerotic burden associated with coronary disease. Although a minor change in creatinine is observed in 85% to 10% of these individuals, a rise sufficient to lead to withdrawal of these agents under trial monitoring conditions occurs in only 1% to 2%. Close follow-up of kidney function indicated that withdrawal of ACE inhibition due to deterioration of renal function was <5% and no greater than placebo.<sup>145</sup> All participants in the CORAL trial were treated with an angiotensin receptor blocker with rare discontinuation.<sup>146</sup> Registry data from the United Kingdom further support the tolerability of renin-angiotensin system blockade in more than 90% of atherosclerotic RVD including more than 75% of patients with bilateral disease.<sup>68</sup> Importantly, there appears to be a mortality benefit in patients treated with these agents.<sup>147</sup>

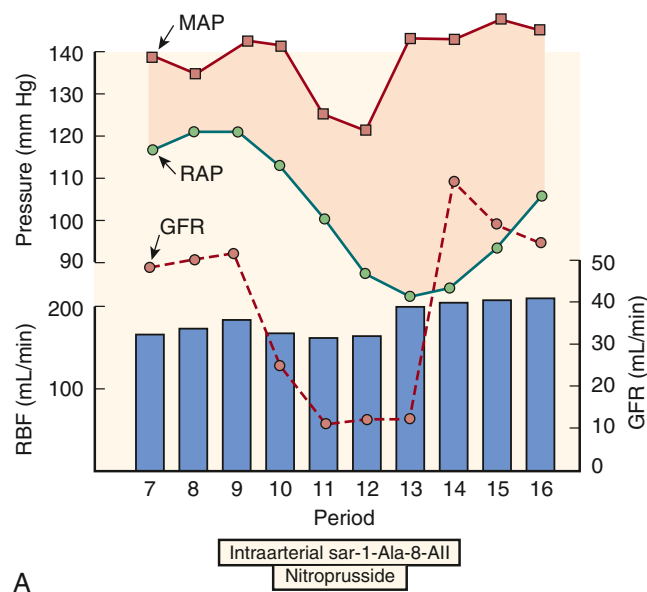
## PROGRESSIVE RENAL ARTERY STENOSIS IN MEDICALLY TREATED PATIENTS

The potential for vascular occlusive disease to progress further is central to long-term management of patients with RVD. Failure to revascularize such kidneys exposes the individual to the hazard of undetected, progressive occlusion, leading to irreversible loss of renal function. Because the treatment trials have been relatively short (2–4 years), this issue remains an important consideration for long-term treatment of atherosclerotic RVD. A firm understanding of the data regarding progressive atherosclerotic disease of the kidney is important for planning both endovascular and surgical revascularization.

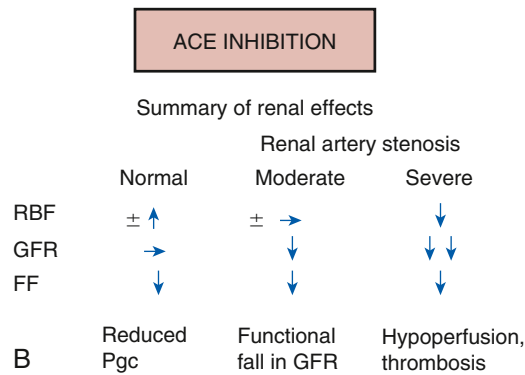
Does progressive renal artery occlusive disease accelerate renovascular hypertension? Moderate anatomic progression alone does not reliably predict functional changes in terms of deteriorating blood pressure control or renal function. Results from prospective Doppler ultrasound studies indicate that a decrement in measured renal size by 1 cm (“renal atrophy”) developed in 5.5% of those with normal initial vessels and 20.8% of those with baseline stenosis >60% during a follow-up interval of 33 months.<sup>148</sup> Changes in serum creatinine were infrequent but did occur in a subset of patients, particularly those with bilateral RVD. Follow-up of the medical treatment arms during short-term studies failed to show major changes in kidney function, although occasional loss of renal perfusion by radionuclide scan was observed.<sup>91</sup>

How often does RVD managed without revascularization lead to clinical progression, either in terms of refractory hypertension or advancing renal insufficiency? Follow-up of patients with incidentally identified RVD identified 69 patients with high-grade renal arterial stenoses (>70%) followed without revascularization for more than 6 months.<sup>149</sup> Their long-term follow-up identified generally satisfactory blood pressure control, although some required more intensive antihypertensive therapy during an average of 36 months’ follow-up. Four of these eventually underwent renal revascularization for refractory hypertension and/or renal dysfunction. Five developed ESRD, of which only one was thought related to RAS directly. Overall, serum creatinine rose from 1.4 mg/dL to 2.0 mg/dL. These data indicate that many such patients can be managed without revascularization for years and that clinical progression leading to urgent revascularization develops in 10% to 14% of such individuals. Measured serum creatinine was higher, and calculated (Cockcroft-Gault) GFR was lower in patients with RVD followed 8 to 10 years, although no patient progressed to ESRD. These observations are consistent with results of prospective trials of medical versus surgical intervention started in the 1980s and extended into the 1990s<sup>150</sup> and more recent RCTs in which both the medical therapy and revascularization arms developed progression to “renal endpoints” in 16% to 22% of cases in CORAL and ASTRAL (Table 47.4).<sup>146,151</sup> No differences in patient survival or renal function were identified. These observations underscore both the relative safety of medical therapy, but also the importance of close follow-up for individuals with identified RVD.

# RENAL HEMODYNAMICS AND FUNCTION DURING SYSTEMIC PRESSURE REDUCTION AND ANGIOTENSIN BLOCKADE IN A DOG WITH RENAL ARTERY CONSTRICTION



A

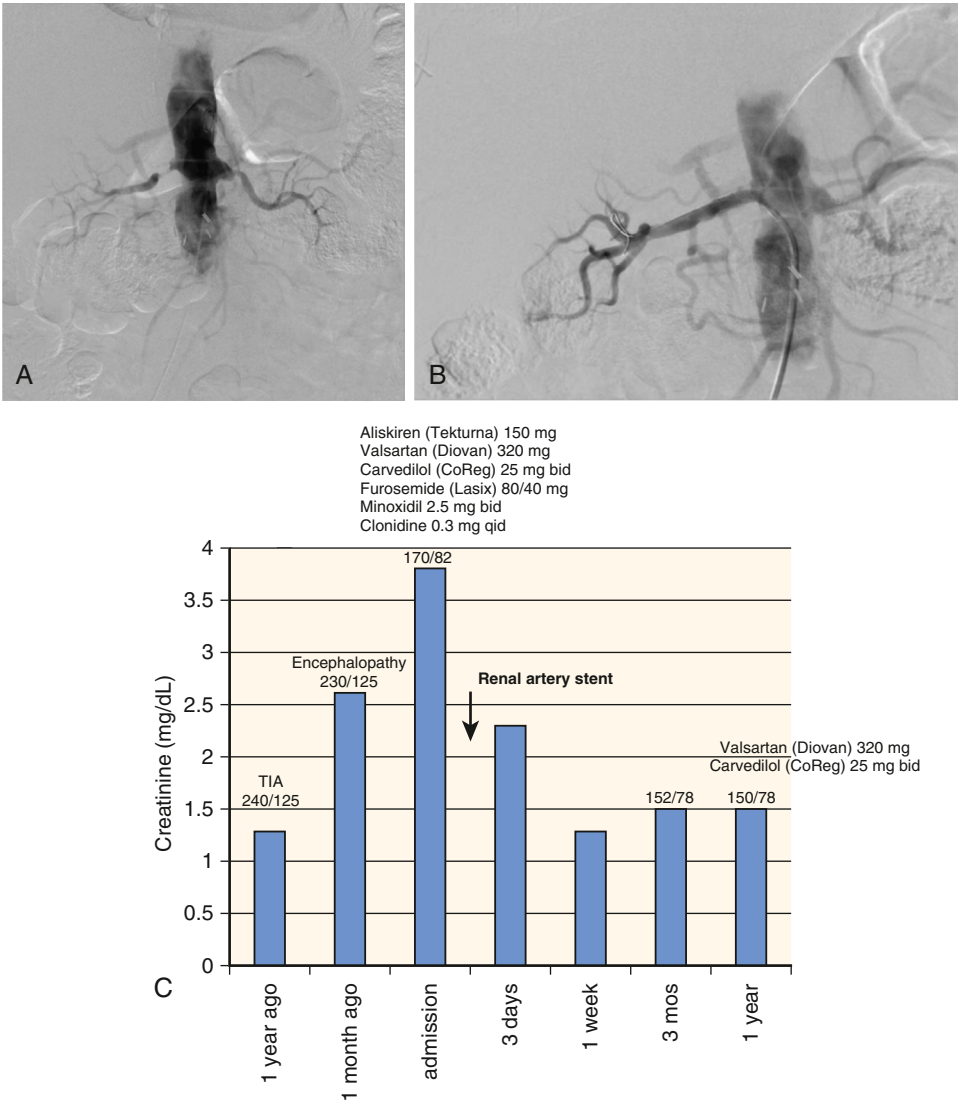


**eFig. 47.12** (A) The glomerular filtration rate (*GFR*) falls beyond a renal artery stenotic lesion during blockade of angiotensin II (produced by intrarenal infusion of sar-1-ala-8-angiotensin II) and pressure reduction induced by sodium nitroprusside. The fall in *GFR* occurs despite preserved renal blood flow (*RBF*), measured by electromagnetic flow probe. These observations illustrate the specific role of angiotensin II in maintaining *GFR* in the poststenotic kidney at reduced perfusion pressures. (B) Summary of the effects of angiotensin-converting enzyme (*ACE*) inhibition and/or angiotensin receptor blockade on *RBF*, *GFR*, and filtration fraction (*FF*) in normal, moderate, and severe levels of renal artery stenosis. As compared with other antihypertensive agents, *ACE* inhibitors and angiotensin receptor blockers (*ARBs*) lead to a fall in *GFR* and filtration fraction due to removal of the efferent arteriolar effects of angiotensin II. When stenosis is sufficiently severe that pressure reduction compromises renal blood flow, the potential for complete occlusion is present, as with other effective antihypertensive agents as well. *MAP*, Mean arterial pressure; *RAP*, renal artery pressure. (From Textor SC. Renal failure related to *ACE* inhibitors. *Semin Nephrol.* 1997;17:67–76.)

While these reports are informative, they leave many questions unanswered. How often does suboptimal blood pressure control of renovascular hypertension accelerate cardiovascular morbidity and mortality? Does one lose the opportunity to reverse hypertension effectively by delaying renal revascularization? These issues require further prospective studies. Most concerning is clinically progressive disease in a subset of “high-risk” patients who present with pulmonary edema, accelerating hypertension and rapidly falling GFR. In such patients, recovery of kidney function and blood pressure control over the long term can be achieved by successful surgical or endovascular restoration of the renal blood supply<sup>73,152,153</sup> (Fig. 47.12).

RENAL REVASCULARIZATION: RENAL ANGIOPLASTY AND STENTING

The ability to restore renal perfusion in high-risk patients with renovascular hypertension and ischemic nephropathy using endovascular methods represents a major advance in this disorder. Restoration of blood flow to the kidney beyond a stenotic lesion intuitively should provide a means to improve renovascular hypertension and halt progressive vascular occlusive injury. Whereas surgical renal artery reconstruction was the mainstay in the 1970s, a major shift in the 1990s led to preferential application of endovascular procedures.<sup>9</sup>



**Fig. 47.12** (A) Renal aortogram illustrating high-grade, bilateral renal arterial lesions in a 73-year-old man who developed accelerated hypertension and loss of glomerular filtration rate associated with transient neurologic deficits a few months before. (B) Aortogram after placement of endovascular stents, illustrating excellent vessel patency and early technical success. This was followed by resolution of his hypertension. (C) Serum creatinine, blood pressure levels, and antihypertensive drug regimen before and after renal artery stent placement. Successful renal artery stenting led to sustained reduction in serum creatinine, improved blood pressure levels, and reduced medication requirements. Such cases reflect “high-risk” subsets that were not well represented in the prospective RCTs. (Modified figure from Textor SC. Attending rounds: a patient with accelerated hypertension and an atrophic kidney. *Clin J Am Soc Nephrol.* 2014;9(6):1117–1123.)

Revascularizing the kidney has both benefits and risks. With older patients developing RVD in the context of preexisting hypertension, the likelihood of a total “cure” for hypertension is small, particularly in atherosclerotic disease. Although complications are not common, they can be catastrophic, including atheroembolic disease, renal infarction, and aortic dissection. Knowing when to pursue renal revascularization is central to effectively managing RVD.

#### ANGIOPLASTY FOR FIBROMUSCULAR DISEASE

Most lesions of medial fibroplasia are located at a distance away from the renal artery ostium. Many of these have multiple webs within the vessel, which can be successfully traversed and opened by balloon angioplasty. Stenting is rarely required unless vessel dissection occurs.<sup>52</sup> Experience in the 1980s indicated more than 94% technical success rates.<sup>154</sup> Some of these lesions (approximately 10%–15%) develop restenosis for which repeat procedures have been used.<sup>155</sup> Clinical benefit regarding blood pressure control has been reported in observational outcome studies in 65% to 75% of patients, although the rates of “cure” are less secure.<sup>54</sup> Cure of hypertension, defined as sustained blood pressure levels <140/90 mm Hg with no antihypertensive medications, may be obtained in 35% to 50% of patients.<sup>156</sup> Predictors of cure (normal arterial pressures without medication at 6 months and beyond after angioplasty) include lower systolic blood pressures, younger age, and shorter duration of hypertension.<sup>156,157</sup>

A large majority of patients with FMD are female at younger ages than those with atherosclerotic disease.<sup>54</sup> In general, such patients have less aortic disease and are at less risk for major complications of angioplasty. Because the risk for major procedural complications is low, most clinicians favor early intervention for hypertensive patients with FMD with the hope of reduced antihypertensive medication requirements after successful angioplasty.

#### ANGIOPLASTY AND STENTING FOR ATHEROSCLEROTIC RENAL ARTERY STENOSIS

After the introduction of percutaneous transluminal renal angioplasty, it was soon evident that ostial lesions commonly failed to respond, in part because of extensive recoil of the plaque, which extended into the main portion of the aorta.<sup>158</sup> These lesions develop restenosis rapidly even after early technical success. Endovascular stents were introduced for ostial lesions in the late 1980s and early 1990s.<sup>159</sup>

The technical advantage of stents is indisputable. An example of successful renal artery stenting is shown in Fig. 47.12. Prospective comparison between angioplasty alone versus angioplasty with stents showed intermediate (6–12 months) vessel patency was 29% and 75%, respectively.<sup>158</sup> Restenosis fell from 48% to 14% in stented patients. As technical success continues to improve, many reports suggest nearly 100% technical success in early vessel patency, although rates of restenosis continue to reach 14% to 25%.<sup>9,160,161</sup>

Regarding outcomes of renal stenting, a systematic analysis under the direction of the Agency for Health Care Research and Quality found that the “strength of evidence” through 2016 regarding benefits and risks of stent revascularization is low, in part, because of major differences between subjects included in prospective RCTs and those reported in observational series.<sup>162</sup> Several professional societies have offered recommendations regarding clinical utilization of revascularization (Table 47.5).<sup>1,163,164</sup> Outcomes are commonly considered in terms of 1. blood pressure control and 2. preservation or salvage of renal function in ischemic nephropathy. Results over 1- to 4-year follow-up are summarized from representative series in Table 47.4 and eTables 47.2 and 47.3 and reviewed in detail elsewhere.<sup>1,165,166,167</sup> Typically, reported falls in blood pressure in observational cohorts are in the range

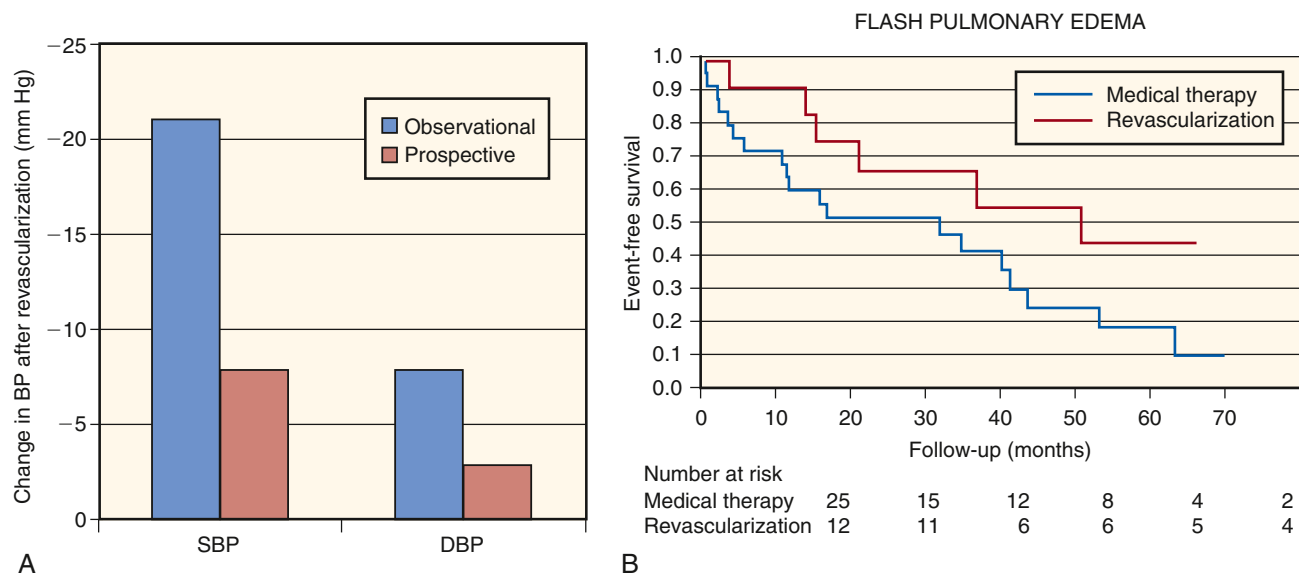
**Table 47.5 Clinical scenarios in which treatment of significant renal artery stenosis may be considered<sup>a</sup>**

| Type of care            | Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appropriate care        | <ul style="list-style-type: none"> <li>Cardiac disturbance syndromes (flash pulmonary edema or acute coronary syndrome [ACS]) with severe hypertension</li> <li>Resistant hypertension (HTN; uncontrolled with failure of maximally tolerated doses of at least 3 antihypertensive agents, 1 of which is a diuretic, or intolerance to medications)</li> <li>Ischemic nephropathy with chronic kidney disease (CKD), with eGFR &lt;45 mL/min/1.73 m<sup>2</sup> and global renal ischemia (unilateral significant RAS with a solitary kidney or bilateral significant RAS) without other explanation</li> </ul> |
| May be appropriate care | <ul style="list-style-type: none"> <li>Unilateral RAS with CKD (eGFR &lt;45 mL/min/1.73 m<sup>2</sup>)</li> <li>Unilateral RAS with prior episodes of congestive heart failure (stage C)</li> <li>Anatomically challenging or high-risk lesion (early bifurcation, small vessel, severe concentric calcification, severe aortic atheroma or mural thrombus)</li> </ul>                                                                                                                                                                                                                                          |
| Rarely appropriate care | <ul style="list-style-type: none"> <li>Unilateral, solitary, or bilateral RAS with controlled BP and normal renal function.</li> <li>Unilateral, solitary, or bilateral RAS with kidney size &lt;7 cm in pole-to-pole length</li> <li>Unilateral, solitary, or bilateral RAS with chronic end-stage kidney disease on hemodialysis &gt;3 months</li> <li>Unilateral, solitary, or bilateral renal artery chronic total occlusion</li> </ul>                                                                                                                                                                     |

<sup>a</sup>Significant RAS is an angiographically moderate lesion (50%–70%) with physiologic confirmation of severity or >70% stenosis.

BP, Blood pressure; eGFR, estimated glomerular filtration rate.

Modified from Parikh SA, Shishehbor MH, Gray BH, et al. SCAI expert consensus statement for renal artery stenting appropriate use. *Catheter Cardiovasc Intervent.* 2014;84:1163–1171.



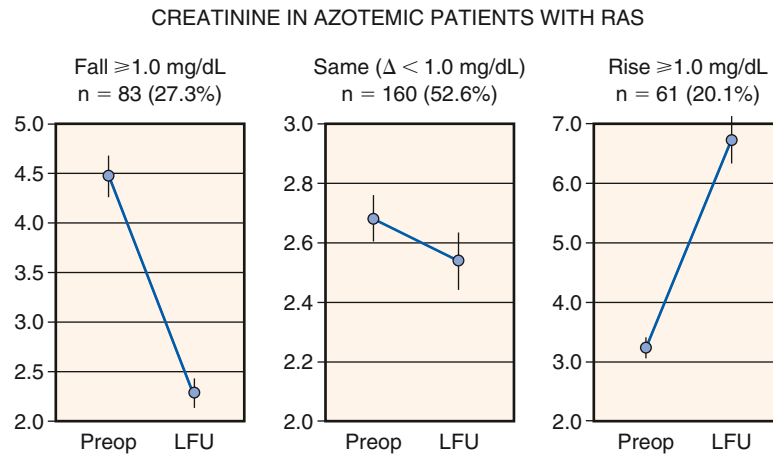
**Fig. 47.13** (A and B) Comparison of blood pressure (BP) changes reported after renal revascularization from a large, observational registry (more than 1000 patients) and from a meta-analysis of three, prospective, randomized trials (210 patients). The large difference between these BP effects is reflected in variable enthusiasm for intervention between clinicians. Initial results from more recent prospective trials (ASTRAL and CORAL) identified only minor differences in achieved BP levels. While results from observational series may overstate treatment benefits, results from prospective trials likely underestimate changes, in part due to limitations in patient recruitment and crossover between treatment arms, ranging from 27% to 44% in early series (see text). (Data from<sup>192,205</sup>) (B) Follow-up data from a registry in the United Kingdom, most of whom were not considered candidates for ASTRAL. This report identified a major mortality risk for patients with renovascular disease and episodes of pulmonary edema that was reduced for patients submitted for renal revascularization. A similar mortality benefit was identified for patients with rapidly progressive renal failure (see text). (From Ritchie and colleagues<sup>73</sup> with permission.)

of 25 to 30 mm Hg systolic, the best predictor of which was the initial systolic blood pressure.<sup>168</sup> Some authors report 42% “improvement” in blood pressure with fewer medications needed, although cures are rare and renal function is generally unchanged.<sup>169</sup> Careful attention to degree of residual patency in 210 patients with stents reported more than 91% patency at 1 year and 79% patency at 5 years.<sup>100</sup> Blood pressures were “cured” or “improved” in more than 80%. In some cases, episodes of angina and recurrent congestive cardiac failure subside.<sup>69,170</sup> Results from prospective RCTs have been less impressive regarding the benefits of stenting.<sup>103,146,151</sup> While these have been controversial insofar as the patients entered in these trials have generally excluded severe or rapidly progressive clinical syndromes, they do demonstrate the limited value of revascularization for moderate disease (Fig. 47.13).

Outcomes regarding recovery of renal function after endovascular revascularization are summarized in **Tables 47.4 and eTables 47.1 and 47.2**. In general, average changes in renal function for atherosclerotic RVD, as reflected by serum creatinine levels, have been small<sup>171</sup> and similar to those observed in azotemic patients after surgical reconstruction.<sup>172,173</sup> However, average group changes in kidney function can be misleading. Careful evaluation of the literature indicates that three distinctly different clinical outcomes are routinely observed. In approximately 27% of cases, revascularization produces clinical improvements in kidney function.<sup>171</sup> For one reported series, mean serum creatinine for this group fell from 4.5 mg/dL to an average of 2.2 mg/dL. There can be no doubt that such patients benefit from the procedure and avoid major morbidity (and probably mortality) associated with advanced renal failure. The bulk of patients, approximating 52%, have little measurable change in

renal function (Fig. 47.14).<sup>171</sup> Whether such patients benefit clinically depends on the true likelihood of progressive renal injury if the stenotic lesion were managed without revascularization, as discussed earlier. Those without much risk of progression gain little, other than potential reversal of hypertrophy in the unaffected kidney. Of greatest concern, however, is the group of patients whose renal function deteriorates further after a revascularization procedure. In most reports, this ranges from 19% to 25% of the patient cohort.<sup>171,174</sup> In some instances, this represents atheroembolic disease or a variety of complications including vessel dissection with thrombosis.<sup>175</sup> Hence nearly 20% of CKD patients face progression of renal insufficiency and the potential of requiring renal replacement therapy.<sup>173,176,177</sup> Predictors of both mortality and ESRD include the level of renal function before intervention and the levels of proteinuria.<sup>79</sup> Importantly, patients with minimal proteinuria enrolled in CORAL had improved mortality outcomes.<sup>5</sup> Deterioration of renal function after stenting sometimes occurs rapidly and may reflect atheroembolic injury, which may be nearly universal after any vascular intervention<sup>178</sup> and acceleration of oxidative stress producing interstitial inflammation and fibrosis.<sup>179,180</sup>

Several series suggest that progression of renal failure attributed to ischemic nephropathy can be reduced by endovascular procedures.<sup>73,174,181</sup> Harden presented reciprocal creatinine plots in 23 (of 32 patients) suggesting that the slope of loss of glomerular filtration rate could be favorably changed after renal artery stenting.<sup>174</sup> Renal function “improved or stabilized” in 69%, indicating that in 31% it worsened, consistent with results reported from other series. Caution must be applied regarding the use of “breakpoint” analysis using reciprocal creatinine plots in this disorder, however. Vascular disease does



**Fig. 47.14 Renal functional outcomes after renal revascularization in 304 azotemic patients (creatinine  $>2.0$  mg/dL) with atherosclerotic renal artery stenosis (RAS).** On average, mean serum creatinine does not change during a follow-up period exceeding 36 months, as observed with most reported series. Group mean values obscure major differences in clinical outcomes, as shown here. Some patients experience major clinical benefit (defined as a fall in serum creatinine of  $>1.0$  mg/dL) (left panel). The largest group has minor changes ( $<1.0$  mg/dL), which might be considered “stabilization” of renal function. The degree of benefit in these patients depends on whether renal function is deteriorating before intervention. The data for the group in the right panel emphasize the failure to observe consistent overall improvement in function because 18% to 22% of patients develop worsening renal function. The exact causes of this deterioration are not clear, although atherosclerotic disease is responsible for a portion. This potential hazard of revascularization must be considered when offering these procedures. *LFU*, Latest follow-up. (From Textor SC, Wilcox CS. Renal artery stenosis: a common, treatable cause of renal failure? *Annu Rev Med.* 2001;52:421–442.)

not affect both kidneys symmetrically and is unlikely to follow a constant course of progression. This can be contrasted with CKD secondary to diabetic nephropathy, for example. As a result, a gradual loss of renal function with subsequent stabilization can be observed equally with unilateral disease leading to total occlusion, as well as successful revascularization. Perhaps the most convincing group data in this regard derives from serial renal functional measurement in 33 patients with high-grade ( $>70\%$ ) stenosis affecting the entire renal mass (bilateral disease or stenosis in a solitary functioning kidney) with creatinine levels between 1.5 and 4.0 mg/dL.<sup>181</sup> Follow-up over a mean of 20 months showed that the slope of GFR loss converted from negative ( $-0.0079$  dL/mg/mo) to positive ( $0.0043$  dL/mg/mo). These studies agree with other observations that long-term survival is reduced in bilateral disease and that the potential for renal dysfunction and accelerated cardiovascular disease risk is highest in such patients.

#### PROSPECTIVE TREATMENT TRIALS COMPARING MEDICAL THERAPY WITH OR WITHOUT ANGIOPLASTY/STENTING

##### Renovascular Hypertension

As noted earlier, many patients with unilateral RAS are managed without restoration of blood flow for a long period, sometimes indefinitely. The judgment about endovascular intervention in a specific case revolves around the anticipated outcome as summarized later. There are now results from several small prospective, randomized trials comparing medical therapy with or without revascularization. The major features of these trials are summarized in Table 47.4 and eTables 47.2 and 47.4 and have been comprehensively reviewed elsewhere. Overall, the results of these trials fail to identify consistent benefits

to adding revascularization regarding BP control or renal function in the enrolled populations. As a result, utilization of renal revascularization has fallen substantially,<sup>2,182</sup> although specific high-risk populations have been identified that benefit clinically.<sup>5,73,183</sup>

##### Complications of Renal Artery Angioplasty and Stenting

Atherosclerotic plaque is complex with heterogeneous calcified, fibrotic, and inflammatory components. Physical expansion of such lesions applies considerable force on the wall and may lead to cracking and release of particulate debris downstream into the bloodstream. A review of 10 published series with 416 stented vessels indicates that significant complications arise in 13% of cases, not counting those that led to the need for dialysis.<sup>9</sup> Complications are listed in Box 47.4 and include hematomas and retroperitoneal bleeding requiring transfusion. Renal function deteriorated in these series on average in 26% of subjects, and 50% (7/14) of those with preprocedure creatinine levels above 400 micromol/L (4.5 mg/dL) progressed to ESKF requiring dialysis.<sup>176</sup> Follow-up data regarding both mortality and ESKF for recent stented cohorts tend to support these observations.<sup>79</sup> Most complications are minor including local hematomas and false aneurysms at the catheter insertion site. Occasionally, severe complications including aortic dissection,<sup>175</sup> stent migration, and vessel occlusion with thrombosis develop.<sup>9</sup> Local renal dissections can be managed by judicious application of additional stents. Mortality related directly to this procedure is small but has been reported in 0.5% to 1.5% of patients.<sup>176,184</sup>

Restenosis remains a significant clinical limitation. Rates vary widely between 13% and 30%, most often developing within the first 6 to 12 months.<sup>160,185,186</sup> Most recent series report 13% to 16% restenosis, sometimes leading to repeat procedures.<sup>160</sup>

### Box 47.4 Complications after percutaneous renal angioplasty and stenting of the renal arteries

#### Minor (most frequently reported)

Groin hematoma  
Puncture site trauma

#### Major (reported in 71/799 treated arteries (9%)<sup>186</sup>

Hemorrhage requiring transfusion  
Femoral artery pseudoaneurysm needing repair  
Brachial artery traumatic injury needing repair  
Renal artery perforation leading to surgical intervention  
**Stent thrombosis:** surgical or antithrombotic intervention

Distal renal artery embolus  
Iliac artery dissection

Segmental renal infarction

**Cholesterol embolism:** renal

Peripheral atheroemboli

Aortic dissection<sup>168</sup>

**Restenosis:** 16% (range: 0%–39%)

**Deterioration of renal function:** 26% (range: 0%–45%)

**Combined complication rate over 24 months:** stenting after failed PTR: 24% (222)

**Mortality** attributed to procedure: 0.5%

**Procedure-related complications:** 51/379 patients in 10 series: 13.5%

**Procedure complications reported from CORAL angiographic core lab (2014):** (159)

Dissection 11/495 (2.2%)

Branch occlusion 6/495 (1.2%)

Distal embolization 6/495 (1.2%)

Single events: wire perforation, vessel rupture, pseudoaneurysm

Total events; 26/495 (5.2%)

### Box 47.5 Surgical procedures applied to reconstruction of the renal artery and/or reversal of renovascular hypertension

Ablative surgery: removal of a “pressor” kidney

Nephrectomy: direct or laparoscopic

Partial nephrectomy

Renal artery reconstruction (require aortic approach)

Renal endarterectomy

Transaortic endarterectomy

Resection and reanastomosis: suitable for focal lesions

Aortorenal bypass graft

Extraanatomic procedures: (may avoid direct manipulation of the aorta)

Require adequate alternate circulation without stenosis at celiac origin

Splenorenal bypass graft

Hepatorenal bypass graft

Autotransplantation with ex-vivo reconstruction

Gastroduodenal, superior mesenteric, iliac-to-renal bypass grafts

Modified from Libertino.<sup>207,208</sup>

## SURGICAL TREATMENT OF RENOVASCULAR HYPERTENSION AND ISCHEMIC NEPHROPATHY

Early experience with RVD was based entirely on surgical intervention, either nephrectomy or vascular reconstruction, with the objective of “surgical curability.”<sup>187</sup> Much of the original data regarding split renal function measurement was geared toward identifying “functionally” significant lesions as a guide for patient selection. Surgical intervention is less commonly performed today and is most often reserved for complex vascular reconstruction and/or failed endovascular procedures.

Methods of surgical intervention have changed over the decades [Box 47.5](#). A review in 1982 emphasized the role of “ablative” techniques including partial nephrectomy.<sup>188</sup> In some cases, nephrectomy of a totally infarcted kidney provides major improvement in blood pressure control at low operative risk. The introduction of laparoscopic surgical techniques makes nephrectomy technically easier in patients for whom vascular reconstruction is not an option. These series reflect widely variable methods of determining blood pressure benefit as discussed later.<sup>125,188</sup>

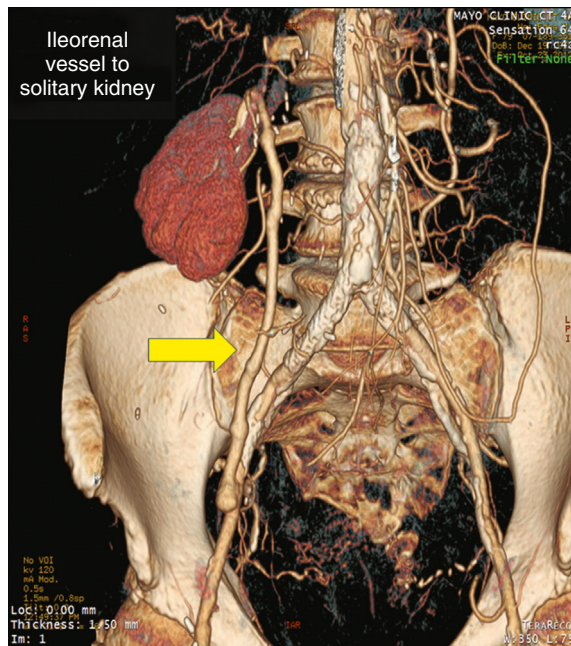
The Cooperative study of RVD in the 1960s and 70s established many of the clinical characteristics of renovascular hypertension<sup>32</sup> and identified some of the limitations and hazards of surgical intervention. They reported mortality

rates of 6.8%, even in excellent institutions. The mean age in this series was 50.5 years. Definitions of operative mortality included events as late as 375 days after the procedure and may have overestimated the hazard. We now appreciate that this is a population of patients with serious comorbidities. Had the authors considered only deaths within the first week, for example, the immediate perioperative mortality was 1.7%.

Subsequent development of improved techniques for patient selection for renal artery bypass and endarterectomy, including prescreening for coronary and carotid disease and for combined aortic and renal artery repair, have altered the practice of vascular surgery.<sup>187,189</sup> Most methods focus on reconstruction of the vascular supply for preservation of nephron mass.

Surgical reconstruction of the renal blood supply usually requires access and manipulation of the aorta. A variety of alternative surgical procedures have been designed to avoid disturbing the badly diseased aorta or when previous surgical procedures make access difficult. These include “extraanatomic” repair of the renal artery using hepatorenal or splenorenal conduits that avoid manipulation of a badly diseased aorta.<sup>190</sup> An example of contemporary surgical reconstruction to protect renal function is illustrated in [eFig. 47.13](#). Results of these procedures have been good in both short- and long-term follow-up studies.<sup>191,192</sup>

The durability of surgical vascular reconstruction is well established. Follow-up studies after 5 and 10 years for all forms of renal artery bypass procedures indicate excellent long-term patency (above 90%) for renal artery procedures alone and when combined with aortic reconstruction.<sup>193</sup> Results of surgery have been good despite increasing age in the reported series. Some studies have compared endovascular intervention (percutaneous transluminal renal angioplasty without stents) and surgical repair. A



**eFig. 47.13** Algorithm summarizing a management scheme for patients with renovascular hypertension and/or ischemic nephropathy. Optimizing antihypertensive and medical therapy for comorbid conditions including dyslipidemia and smoking is paramount to reducing cardiovascular morbidity and mortality in atherosclerotic disease. Decisions regarding the timing of renal revascularization procedures depend on the clinical manifestations (see text) and whether blood pressure and kidney function remain stable. *ACE*, Angiotensin-converting enzyme; *GFR*, glomerular filtration rate; *PTRA*, percutaneous renal angioplasty; *RAS*, renal artery stenosis.

prospective comparison of endovascular stents compared with open surgical renal revascularization suggested that patency over 4 years was better with open surgical repair but overall the outcome of the two procedures did not differ.<sup>194</sup>

Studies in patients with bilateral renal artery lesions or vascular occlusion to the entire renal mass indicate that restoration of blood flow can lead to preservation of renal function in some cases.<sup>195</sup> Most often, this has been undertaken when evidence for preserved or collateral blood supply, sometimes from capsular vessels, is evident by renography. Occasionally, revascularization can lead to functional recovery sufficient to eliminate the need for dialysis.

## PREDICTORS OF LIKELY BENEFIT REGARDING RENAL REVASCLARIZATION

Identification of patients most likely to improve blood pressure and/or renal function after renal revascularization remains elusive. As noted earlier, functional tests of renin release, such as measurement of renal vein renin levels, have not performed universally well. Many of these studies are most useful when positive. Stated in other terms, the likelihood of benefit improves with more evident lateralization, yet evidence indicates relatively poor negative predictive value. Specific criteria to establish the severity of vascular compromise do not reliably predict the response of clinical syndromes. Recent progression of clinical hypertension, deterioration of renal function, and/or pulmonary edema remain among the most consistent predictors of improved blood pressure after intervention.

Predicting favorable renal functional outcomes is also difficult. Both surgical and endovascular procedures are least likely to benefit those with advanced renal insufficiency, usually characterized by serum creatinine above 3.0 mg/dL. Nonetheless, occasionally patients with recent progression to advanced renal dysfunction can recover GFR with durable improvement over many years. Small kidneys, as identified by length <8 cm, are less likely to recover function, particularly when little function can be identified on radionuclide renography.<sup>196</sup> Reports of renal resistance index measured by Doppler ultrasound in 5950 patients indicate that identification of lower resistance was a favorable marker for both improvement in GFR and blood pressure, whereas elevated resistance index was an independent marker of poor outcomes.<sup>106</sup> None of these associations are absolute, and more recent studies identify favorable outcomes in some patients with adverse predictors.<sup>184</sup> As noted earlier, outcomes for patients with low urinary albumin/creatinine ratio are distinctly better than for those with higher levels.<sup>5,79</sup> Recent deterioration of kidney function or hypertension portends more likely improvement with revascularization.

## FUTURE RESEARCH IN RENOVASCULAR DISEASE

How best to identify and recover filtration in poststenotic kidneys remains a challenging area of investigation. Measurement of renal vein cytokines indicates that poststenotic kidneys exhibit abundant proinflammatory signaling, even with relatively preserved oxygenation.<sup>180</sup> Studies of human kidneys undergoing temporary arterial occlusion indicate that mitochondrial swelling and disruption develop rapidly.<sup>197</sup> These data support experimental models showing increased oxidative stress and mitochondrial

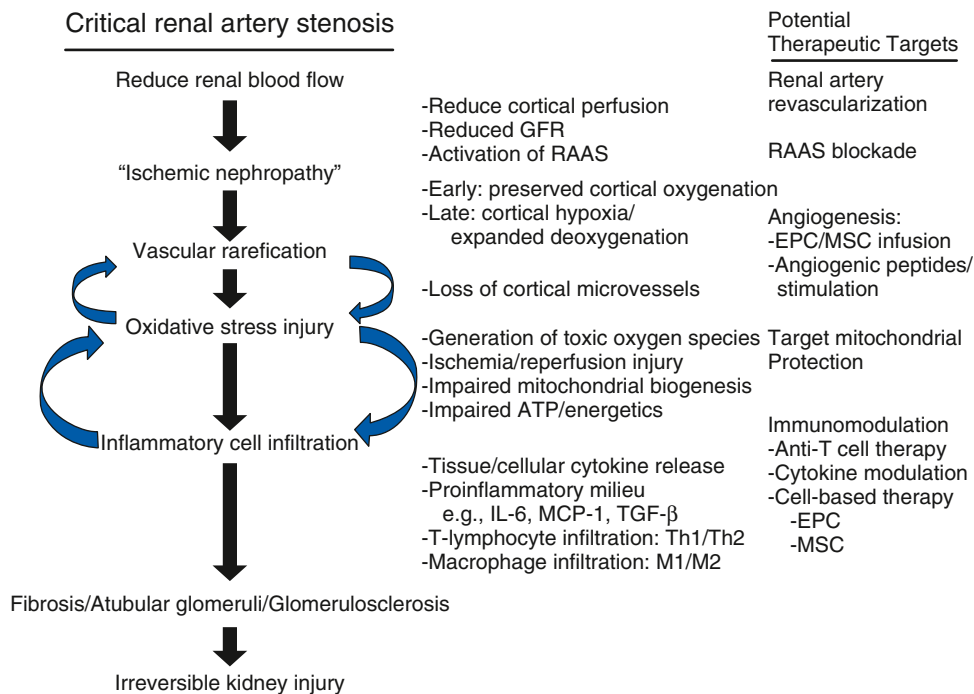
dysfunction under conditions of chronically reduced renal blood flow.<sup>198,199</sup> Protection of mitochondrial structure and function using infused agents that stabilize cardiolipin and inhibit opening of the mitochondrial transition pore appears to facilitate microvascular recovery and kidney regeneration after renal revascularization.<sup>49</sup> Preliminary studies in human subjects support this approach.<sup>200</sup>

Newer studies emphasize the role of inflammatory pathways and accumulation of activated T cells and macrophages in vessels and tissue of the atherosclerotic kidney.<sup>31</sup> Experimental models utilizing infused endothelial progenitor cells<sup>44</sup> and mesenchymal stem cells<sup>201</sup> respond to revascularization with improved microvascular regeneration. Initial studies in human atherosclerotic RAS using autologous mesenchymal stem cells delivered into the renal artery suggest that cortical blood flow and tissue oxygenation increase over a 3-month period.<sup>202,203</sup>

Recognition of the role and interactions between these pathways offers the possibility for targeted therapy in patients with occlusive RVD. Some of these are highlighted in Fig. 47.15.

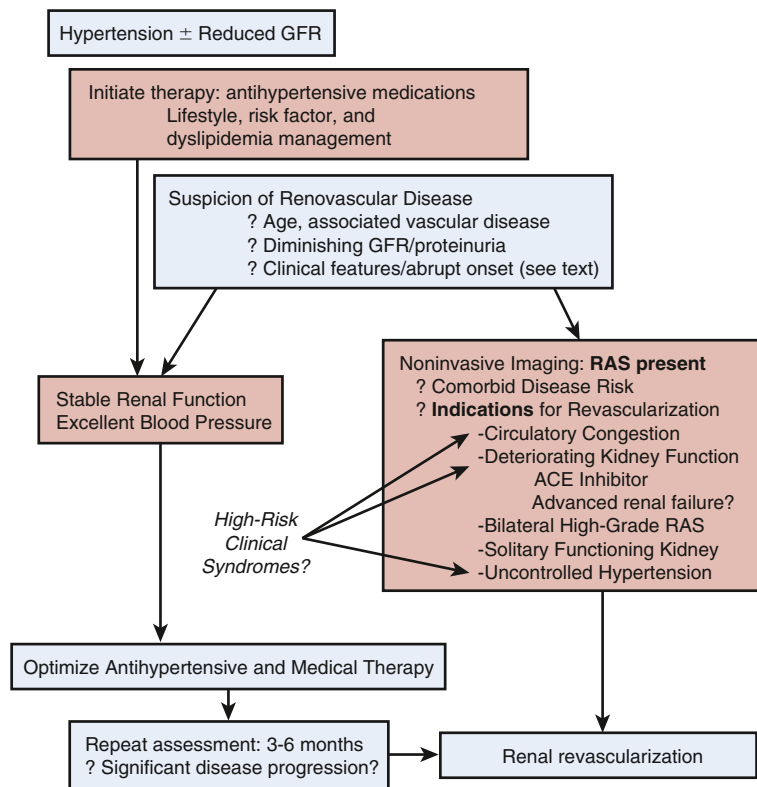
## SUMMARY

RVD is common, particularly in older subjects with other atherosclerotic disease. It can have multiple clinical consequences, ranging from asymptomatic “incidentally discovered” disease to accelerated hypertension and progressive renal failure. With improved imaging and increasing patient age, significant renal artery disease is detected more frequently. The clinician must evaluate both the impact of the renal artery disease in the individual patient and the potential risk/benefit ratio for renal revascularization. Many patients with fibromuscular dysplasia and other disorders benefit from restoring renal artery patency in terms of blood pressure control. An algorithm to guide treatment and reevaluation of patients with atherosclerotic RAS is presented in Fig. 47.16. Application of this strategy relies heavily on considering comorbid risks and the evolution of both blood pressure control and kidney function over time. Management of cardiovascular risk and hypertension are the primary objectives of medical therapy. For most patients, the realistic goals of renal revascularization are to reduce blood pressure medication requirements and stabilize renal function over time. Patients with bilateral disease or stenosis in a solitary functioning kidney may have lower risk of circulatory congestion (“flash” pulmonary edema or its equivalent) and lower risk for advancing renal failure after revascularization of the kidney. It is essential to appreciate the risks inherent in either surgical or endovascular manipulation of the diseased aorta. Hence the decision to undertake these procedures should include consideration of whether the potential gain warrants such risks. In many cases, improved blood pressure and recovery of renal function justify the costs and hazards. Long-term follow-up of both blood pressure and renal function is important in all cases whether revascularization is performed or not, particularly because of the potential for disease progression, restenosis, and/or recurrent disease. Optimal selection and timing for medical management and revascularization depend largely on the comorbid conditions of each patient.



**Fig. 47.15 Injury pathways and targets in renovascular disease.** Working diagram highlighting recent experimental studies delineating specific pathways of oxidative stress injury and inflammatory injury pathways in the poststenotic kidney. The right panel identifies specific therapeutic targets that may alleviate these injury pathways, over and above simply restoring blood flow. Many of these pathways likely overlap in many forms of kidney injury. *EPC*, Endothelial progenitor cells; *MCP*, monocyte chemoattractant protein. (From Textor SC, Lerman LO. Paradigm shifts in atherosclerotic renovascular disease: where are we now? *J Am Soc Nephrol*. 2015;26(9):2074–2080 with permission.)

#### MANAGEMENT OF RENOVASCULAR HYPERTENSION AND ISCHEMIC NEPHROPATHY



**Fig. 47.16 Algorithm summarizing a management scheme for patients with renovascular hypertension and/or ischemic nephropathy.** Optimizing antihypertensive and medical therapy for comorbid conditions including dyslipidemia and smoking are paramount to reducing cardiovascular morbidity and mortality in atherosclerotic disease. Decisions regarding timing of renal revascularization procedures depend on the clinical manifestations (see text) and whether blood pressure and kidney function remain stable. *ACE*, Angiotensin-converting enzyme; *GFR*, glomerular filtration rate; *PTRA*, percutaneous renal angioplasty; *RAS*, renal artery stenosis.

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